Finance Gets Personal: The Lasting Effects of Idiosyncratic Crisis Experiences on Investment Behavior *

Hamid Boustanifar† Hao Jiang‡ Dan Zhang§ Lu Zheng¶

November, 2025

Abstract

Using Norwegian administrative data, we examine how idiosyncratic personal experiences during an aggregate market shock shape long-run investment behavior. We find that investors who suffered large losses during the 2008 financial crisis persistently reduced their equity allocations and became more likely to exit the stock market entirely. Among those who remained invested, crisis experiences led to significantly more concentrated portfolios—an unintended consequence that increased idiosyncratic risk despite investors' apparent desire to reduce risk exposure. These effects persist for at least five years after the crisis, even as aggregate market conditions recovered and model-based expected returns remained high. We also show that personal crisis experiences affect equity allocation decisions even after controlling for formative year experiences. Moreover, positive formative-year stock market returns partially offset crisis-driven reduction in stock allocation, demonstrating how both direct experience of personal losses and indirect experience of early-life equity market conditions jointly drive persistent investor heterogeneity.

Keywords: Individual investors, portfolio allocation, stock market participation, crisis experience, household finance, financial crisis

JEL Codes: D14, G10, G11

*The authors wish to thank Angela Ma and seminar participants at EDHEC Business School, Erasmus University Rotterdam, Norges Bank, Norwegian University of Science and Technology, Nord University, Oslo Metropolitan University, and Vrije Universiteit Amsterdam for feedback and discussions. Dan Zhang and Hamid Boustanifar acknowledge financial support from the Research Council of Norway through a Finansmarkedsfondet grant (Project Number 328618)

†EDHEC Business School. hamid.boustanifar@edhec.edu

‡Michigan State University. jiangh@broad.msu.edu

§Oslo Business School, Oslo Metropolitan University. danzhang@oslomet.no

¶Paul Merage School of Management, UC Irvine. luzheng@uci.edu

"The world is not what I think, but what I live through."

— Maurice Merleau-Ponty

1 Introduction

Academic studies in investments have traditionally focused on security prices and other aggregate behaviors, such as flows into assets and trading volume. Although behavioral finance explores the psychology behind financial decisions, it often views investors as general groups such as "retail investors", aiming to identify their common biases and study their average behavior. Such a focus on the aggregate, while useful for macro understanding, overlooks individual differences. In contrast to traditional finance that assumes investors are fully rational and maximize their wealth or utility, at the personal level, each investor's decisions may be shaped not only by market signals but also by their idiosyncratic personal histories and experiences.

The recent literature provides valuable insight on this issue by showing how past *market* experiences influence investor behavior. For instance, Malmendier and Nagel (2011) and subsequent studies by Malmendier et al. (2020a) and Malmendier et al. (2020b) show how aggregate market shocks influence the investment behaviors of different age cohorts, such as more conservative financial strategies among those who experienced lower stock market returns during their formative years. However, while these analyses provide valuable insight into the effects of past market conditions on cohort behavior, they overlook the granularity of diverse individual experiences within these broad groups. As Malmendier (2021) points out, the study of *personal* experiences is of interest, because it captures effects more deeply encoded at the personal level and opens a window to analyze a richer set of experiences. Our study aims to fill this gap by focusing on how unique and specific personal experiences during the 2008 financial crisis affect subsequent investment decisions in the long-run.

Neoclassical finance posits that all investors, regardless of their past losses or experiences, evaluate future investment opportunities based solely on the potential return and risk, aiming to maximize wealth or utility. This rational evaluation suggests that investment decisions post-crisis should be based on updated market information rather than historical experiences. Motivated by the literature on market experience, our alternative hypothesis posits that unique personal experiences during financial

crises significantly affect subsequent investment behavior, leading to deviations from rational models.

In this paper, we examine how individual experiences during the 2008 financial crisis—an aggregate shock whose timing and severity were largely exogenous to individual investors—shaped subsequent investment decisions. This setting provides a compelling opportunity to study the long-term behavioral effects of personally experienced losses during a major market disruption. Our analysis leverages comprehensive data on the entire population of Norwegian investors, including both end-of-year holdings and daily stock transaction records from 2007 to 2013. We merge these data with rich demographic and socioeconomic information. For each investor, we compute the loss in the equity portfolio during the crisis—capturing both unrealized ("paper") losses and realized losses, made possible by the daily transaction data—expressed as a percentage of the individual's total personal wealth at the end of 2007.

We find that investors who experienced a large loss in the financial crisis tend to reduce their stock allocation for up to five years after the crisis. For instance, an experience of 33% (the 90th percentile) or a greater loss of wealth resulting from equity investments during 2008 is associated with a decrease in equity allocation by 3.4% from 2008 to 2009, by 4.6% from 2008 to 2011 and by 4.4% from 2008 to 2013. This impact of crisis loss is robust to controlling for a comprehensive set of pre-crisis individual characteristics such as personal wealth, liabilities, various sources of income, ownership of private firms, employment status, age, marital status, and education. It is also robust to alternative quantification of large losses in the financial crisis.[1]

Our estimates also show that the probability that an investor exits the stock market (extensive margin) after a large loss in equity investment during the financial crisis increases from 3.7% to 7.0% in the first year, from 12.4% to 20.5% in the three-year period, and from 19.5% to 30.1% in the five-year period. These results demonstrate that the experience of equity loss in a financial crisis has a prolonged impact on an investor's willingness to take equity market risk.

Conditional on remaining in the stock market, we find that investors who experienced large losses hold significantly more concentrated portfolios (intensive margin). Measured by the Herfindahl-Hirschman Index (HHI), portfolio concentration

[1]During our sample period, there was no liquidation or bankruptcy in listed Norwegian banks. Therefore, our results do not mechanically reflect possible bank failures.

increases on average by 1.6% from 2008 to 2009, 4.7% to 2011, and 5.9% to 2013 for investors with losses at or above the 90th percentile. Consistent with this pattern, these investors also reduce the number of unique stocks in their portfolios by 5.8% from 2008 to 2009, by 11.7% to 2010, and by 15.5% to 2013. Importantly, these concentration effects are not mechanical consequences of portfolio liquidation—they reflect deliberate rebalancing decisions by continuing investors. The results reveal an important tension: while investors respond to crisis losses by reducing their equity exposure, those who remain invested inadvertently increase their exposure to idiosyncratic risk through greater concentration, potentially undermining their desire to reduce risk.

To address the concern that heterogeneous personal experiences during the financial crisis may reflect unobserved individual characteristics, we construct a sample of sibling pairs. Prior research shows that a substantial share of cross-sectional differences in financial decision making is related to family background and genetic factors (Cesarini et al., 2009, 2010). By comparing siblings, we hold constant many unobservable traits shared within families—such as upbringing, socioeconomic environment, and stable preference components—while also controlling for observable pre-crisis characteristics. Within these sibling pairs, we find that the sibling who experienced a larger equity loss in 2008 subsequently allocates a smaller share of financial assets to stocks, is more likely to exit the stock market, and, if remaining invested, holds fewer individual stocks with higher portfolio concentration relative to their sibling who experienced a smaller loss. In short, our main findings persist even within families, providing strong evidence that the effects we document are not driven by unobserved family-level or genetic differences.

To understand the impact of the large loss experience during the financial crisis on an investor's stock allocation decision, we exploit insights from a standard model of risky asset allocation (Campbell and Viceira (2002), Chapter 6), in which an investor's risky asset allocation depends on three key parameters: expected excess return on the risky asset, volatility risk of the risky asset, and (relative) risk aversion. With the observed volatility risk and the assumption of a constant relative reversion (following Campbell and Viceira (2002), we can mute the last two parameters and consider the first, namely the change in expected excess return on the risky asset. For each individual investor in our sample, we compute a measure of expectations of future stock market returns using statistical estimates from Greenwood and Shleifer (2014). We find that for many individual investors experiencing large losses in the 2008 crisis, their

subsequent expectations of stock market return in excess of the risk-free rate are negative. We argue that the negative expectation of stock market risk premia can explain their reduced stock market allocations and increased tendency to exit the stock market after the crisis when increased stock market allocations are more attractive.

Building on insights from the experience effect literature, we next study how formative year experiences affect investment behavior. For each investor in our sample, we construct a variable on the formative year experience, representing the cumulative returns of the stock market during their formative years, from ages 11 to 20, in the spirit of Malmendier and Nagel (2011). We first show that our previous findings on the impact of personal crisis experience are robust to controlling for formative year experience. Second, we find that the performance of the stock market during the formative years of an investor has an impact on their equity allocation decision alongside personal crisis experience: among those who experienced large losses during the crisis, a high formative year stock market return is associated with a greater increase in stock allocation and stock concentration. This result supports the thesis of Malmendier and Nagel (2011) on the importance of investor experiences in financial risk taking. In addition, we show that our previous findings that individual investors experiencing a large loss in the financial crisis tend to reduce their stock allocation and increase their stock concentration are robust to controlling for formative year experience. Finally, when the 2008 crisis experience resonates with the investor's formative period experience, the negative impact on risk taking intensifies. That is, the interaction between the direct experiences of investors about the loss of their 2008 equity portfolio and their indirect experiences of the loss of the equity market when they grew up together shapes their investment decisions.

Our paper contributes to the literature on how investor experiences shape financial decision making.[2] We focus on the effects of *individual*—rather than common or economy-wide—experiences on both short-term and *long-run* portfolio decisions. Most prior work studies economy-wide or formative experiences, such as living through periods of high stock market returns or high inflation (Malmendier and Nagel, 2011, 2016).[3] Our work complements this literature by studying how *personal* experiences during financial crises affect *changes* in the portfolio allocations for the same investors over time. Moreover, our

[2]See Malmendier (2021) for an excellent survey.

[3]An exception is Knüpfer et al. (2017) that investigate the effect of regional labor market shocks (that vary within cohorts) on future stock market participation.

study provides evidence on how formative experiences interact with subsequent personal experiences in shaping financial decisions.

While an extensive literature has examined the causes and consequences of the 2008–2009 financial crisis for stock and housing markets, financial institutions, and institutional investors, only a limited number of studies have investigated how individual investors behave during and after financial crisis (Hoffmann et al., 2013; Bateman et al., 2011; Bucher-Koenen and Ziegelmeyer, 2014; Guiso et al., 2018; Andersen et al., 2019; Hoopes et al., 2022). These studies focus either on behavior during the crisis or immediately afterwards. However, it is not clear that such effects persist in the long run. In many economic settings, households gradually reverse initial reactions as conditions normalize, wealth is rebuilt, or beliefs adjust. If crisis-induced exits from the stock market are similarly transitory, their long-term welfare implications would be limited. This makes it essential to distinguish temporary responses from persistent behavioral changes—an issue our study directly addresses by examining both the immediate and long-lasting effects of crisis experiences on investor behavior. In addition, we link this work to the literature on formative experiences by studying how long-run macroeconomic experiences interact with personal experiences during the financial crisis.[4]

Finally, a large literature examines the stock market participation puzzle—the fact that many households do not hold equity despite predictions from standard portfolio theory (e.g., Mankiw and Zeldes, 1991; Haliassos and Bertaut, 1995; Campbell, 2006). Prior work highlights a range of determinants, including wealth, income, education, health, risk tolerance, cognitive skills, and social interaction.[5] Empirical studies typically compare participants with non-participants (Hong et al., 2004; Calvet et al., 2007; Guiso et al., 2018; Choi and Robertson, 2020) or examine changes in participation following specific shocks (Brown et al., 2008; Giannetti and Wang, 2016). Far fewer studies investigate why investors exit the stock market. Notable exceptions include Giannetti and Wang (2016), who show that participation falls after corporate fraud revelations, and Fagereng et al. (2017), who demonstrate in a life-cycle model that

[4]Moreover, as Andersen et al. (2019), which is closest paper to ours, note, their measurement of experience is constrained by end-of-year holdings, which cannot capture intra-year trading or the actual losses realized when shocks occur. In contrast, we use comprehensive transaction-level data for the full population of investors, allowing us to reconstruct the precise timing and magnitude of realized and unrealized losses and to observe all buy and sell decisions as they unfold.

[5]See Campbell (2006) and Guiso and Sodini (2013) for reviews.

investors exit upon retirement. Our study adds a new perspective by showing that personal experiences during the 2008 financial crisis play a central role in investors' decisions to exit the stock market, with effects that persist for many years.

In general, our findings help us to better understand the experiential and psychological foundations of belief formation. In his 2025 AFA lecture, Barberis highlights the need to explore how personal experience shapes investor beliefs, particularly under conditions of uncertainty and salience.[6] Our results provide empirical support for this view, showing that even in the presence of improving aggregate signals and favorable model-based expected returns, investors' personal crisis experiences can lead to persistently different beliefs and disengagement from the stock market in the long-term.

The paper proceeds as follows. In the next section, we describe the construction of our sample and the data set. Section 3 presents our main results on changes in stock allocation and stock concentration following the personal crisis experience in 2008. In Section 4, we add the formative year experience in our baseline models and discuss its interaction with the experience of personal crisis. Section 5 provides robustness checks, and Section 6 presents our conclusions.

2 Data and sample

Our analysis is based on administrative register datasets from Norway that offer several advantages for studying how individual experiences during the 2008 financial crisis shaped subsequent investment decisions.

2.1 Institutional setting

Norway is a wealthy, developed economy with a population of 5.5 million, and the Oslo Stock Exchange represents a medium-sized European market operating under EU-based regulations. Norwegian banks remained relatively stable during the 2007–2009 financial crisis due to strict financial regulations and conservative banking practices, providing a setting where the crisis represented a significant market shock

[6] 2025 AFA Lecture - Behavioral Finance at 40: Progress, Open Questions, and New Directions by Nicholas Barberis. https://afajof.org/2025-annual-meeting-and-panel-sessions/

without systemic collapse.

The Norwegian data environment is particularly well-suited for our research question. Norway maintains comprehensive administrative records on the entire population through Statistics Norway (Statistisk sentralbyrå), which collects data directly from the Norwegian Tax Agency, welfare agencies, employers, and financial institutions to administer Norway's wealth tax. This wealth tax is levied on global net worth as of December 31st each calendar year. The data cover detailed information on income (employment, business, and capital income), wealth (bank deposits, stocks, mutual funds, bonds, shares in private firms, real property), and sociodemographic characteristics (gender, age, marital status, educational attainment). Because wealth- and income-related items are pre-filled in tax returns with individuals only reporting corrections, the data are highly reliable.

Crucially for our analysis, we can link these population-wide annual records with daily transaction data for all stock trades on the Oslo Stock Exchange by individual investors from 2007 to 2013. This combination provides three key advantages. First, we observe each investor's total personal wealth at the end of 2007, which serves as the pre-crisis measure. Second, the daily transaction records enable us to calculate both realized and unrealized losses during the crisis period with precision. Third, we can trace how crisis experiences shaped subsequent investment behavior over multiple years using the same comprehensive population. The granularity and completeness of these linked datasets allow us to construct individual-level measures of financial crisis exposure that would be difficult or impossible to obtain in other settings.

2.2 Data sources

Our data cover all common stock transactions on the Oslo Stock Exchange by Norwegian residents between 2007 and 2013, along with the sociodemographic characteristics of each investor. The data come from three sources, linked via unique identifiers for individuals and for stocks.

First, to characterize individuals' stock trading behaviors, we collect information from the Norwegian Central Securities Depository (Verdipapirsentralen or VPS in Norwegian), which is the central securities depository in Norway that records all stock transactions on the Oslo Stock Exchange, the only regulated market for securities

trading in Norway. We obtain all common stock transactions on the Oslo Stock Exchange from all Norwegian residents from January 2007 to December 2013. For each transaction, we have information on the date of the transaction, the international securities identification number (ISIN) of the stock, the type of the transaction, the transaction price, the number of shares purchased or sold, the transaction value, and an anonymized identifier of the individual that can link individuals in other data sources.[7]

Second, we obtained daily stock prices, trading volumes, and market index data from the Oslo Stock Exchange. In addition, we collected the end-of-year holdings (at the 31 December each year) of each stock for each individual in the VPS. Using these holdings and transactions data, we can construct the stock portfolio of each individual at any point during our sample period. For those individuals who made multiple transactions of the same stock on the same day, we aggregated the transactions at the individual-stock-date level. In such a case, we can also capture returns within the trading day as we have information on the prices bought or sold for every transaction.

Third, we collect sociodemographic characteristics of investors from the Norwegian government statistics registry, Statistics Norway. This data includes annual records of income variables (e.g., income from employment, business, and capital income), wealth variables (e.g. bank deposits, stocks, mutual funds, bonds and money market funds, shares in private firms, cars, and real property), age, marital status, education, gender, along with the anonymized identifiers that allow us to match the sociodemographic variables with the transaction data.

2.3 Sample and variables

2.3.1 Crisis period

To capture the personal experience with the stock market crisis, we first need to identify the crisis during our sample period. The global financial crisis in 2008 had a significant effect on the Norwegian stock market. As shown in Figure 1 (Panel A), the cumulative returns, calculated using the monthly return of Oslo Børs All-Share Index, increased steadily from 2003 to 2007 before plummeting from about 490% in December

[7]We strictly follow the General Data Protection Regulation (GDPR) and make sure that the data is anonymous and not directly identifiable. To preserve anonymity, we do not have the real social security numbers. Instead, Statistics Norway provided us with an anonymized version of pseudo social security numbers that allow us to match the data across different sources.

2007 to approximately 220% in December 2008, representing a decline of almost 60%. The most significant drop occurred in September and October 2008, with monthly market returns of -24% and -22%, respectively. The maximum drawdown, measured by the difference between the highest peak and the lowest trough values of Oslo Børs All-Share Index, surged to nearly -60% at the end of 2008. The realized volatility, based on the monthly volatility of Oslo Børs All-Share Index, peaked in October 2008.

[Insert Figure 1 about here]

In Panel B of Figure 1, we plot the implied volatility index for the Norwegian equity market, NOVIX, in logs and levels, at a monthly frequency from January 2007 to December 2013.[8] Until mid-2008, NOVIX remained low compared to the levels observed later during the crisis. However, NOVIX surged dramatically from about 28% at the end of August to 62% at the end of October, then began to decrease. [9] These patterns collectively indicate a significant market upheaval in 2008, which we will refer to as the crisis year in our analysis.[10]

2.3.2 Allocation in stocks and other financial assets

To study the effect of individual experiences on investment decisions, we start by looking at changes in stock allocation over the one, three, and five-year horizons. Specifically, the n-year change in stock allocation is defined as:

$$\Delta StockAllocation_i(n) = 100 \times \frac{\sum_{m=1}^{M} \Big[Holding_{m,i,2008+n} - Holding_{m,i,2008}\big(1 + Return_m(n)\big) \Big]}{FinancialAssets_{i,2007}} \quad (1)$$

where $n = 1$, 3, or 5, $Return_m(n)$ is the n-year stock return on stock m; and M is the number of stocks held in the investor i's portfolio. In other words, the term

[8] We are grateful to Haakon Guttormsen for sharing the data on NOVIX, as constructed and evaluated in Bugge et al. (2016).

[9] The data at a daily frequency reveals that NOVIX started to increase drastically from mid-September 2008 and peaked at 80% at the beginning of October.

[10] Hoopes et al. (2022) study how the spike in the VIX interacts with investor characteristics to drive their selling decisions. As shown in their study, the VIX spiked in 2008 and descended in 2009. Similar to the US market, the turmoil in the Norwegian stock market abated in 2009; therefore, our post-crisis period starts from 2009.

$Holding_{m,i,2008}\big(1 + Return_m(n)\big)$ represents the buy-and-hold value of the stocks held in 2008. Our measure of stock allocation is the difference between the actual portfolio value and the buy-and-hold value of the stocks held in 2008, scaled by the value of financial assets at the end of year 2007. This measure captures the active trading behavior of an investor by removing the impact of a market-wise negative shock on the portfolio value from passive holding. The years one, three, and five after the crisis refer to the year 2009, 2011, 2013. We use the value of financial assets at the end of 2007 as the denominator to ensure that it was not affected by the crisis yet. Again, personal financial assets including stocks portfolio, as well as bank deposits, mutual funds, bonds, and money market funds, and unlisted securities. Note that all values in Norwegian krone were inflation adjusted using the Consumer Price Index (CPI).

Next, we capture changes in investors' stock concentration using three measures, namely stock liquidation, change in the number of stocks, and a concentration index. *Liquidation* is a dummy variable that equals one if the investor liquidates all stocks and zero otherwise. This measure is defined for full sample—all investors who held stocks at the end of 2008.

For investors who remain in the stock market (i.e., continue to hold at least one stock), we measure portfolio concentration changes using two additional variables. *Change in Number of Stocks* is the difference between the number of unique stocks held at the end of this period and that of the previous period. *Change in concentration index* (ΔHHI) is the change in *HHI* specified as follows, where *HHI* measures the sum of the squared weights of individual stocks in the portfolio:

$$\Delta HHI_i(n) = 100 \times \big(\sum_{m=1}^{M} S^2_{m,i,2008+n} - \sum_{m=1}^{M} S^2_{m,i,2008} \big) \tag{2}$$

where $s_{i,m,t}$ is the weight of stock m in investor i's portfolio at time t. In other words, the n-year change in stock concentration (where n is 1, 3, or 5), is measured as the difference between the *HHI* index n years after the crisis and the *HHI* in 2008. *HHI* captures portfolio concentration by measuring the sum of the squared weights of individual stocks in the portfolio. A positive $\Delta HHI_i(n)$ indicates an increase in stock concentration or a decrease in stock diversification. We scale the difference with 100 to facilitate the reading of the coefficient in the regression analysis. In our sample, the average *HHI* at the end of 2008 is 80 with a median of 100, indicating that many investors only held one stock at the time.

In addition to stock allocation and concentration, we also have data from the tax authority on the end-of-year value of cash deposits, mutual funds, bonds, money market funds, and other unlisted securities. Together with stocks, these are the items that add up to personal financial assets. Unlike stocks, we do not have detailed information on transactions and holdings information for cash deposits, mutual funds, bonds, money market funds, and other unlisted securities. Thus, we can only measure changes in the allocation of these asset classes based on their end-of-year values. Consequently, we cannot distinguish between active changes in allocation and passive changes resulting from changes in the market value of these assets. The estimation equation is given below:

$$\Delta AssetsAllocation_i(n) = 100 \times \frac{Holding_{m,i,2008+n} - Holding_{m,i,2008}}{FinancialAssets_{i,2007}} \tag{3}$$

where $n = 1$, 3, or 5, $\Delta AssetsAllocation_i(n)$ is a set of variables that includes changes in bank deposits, in mutual funds, in bonds and money market funds, and in unlisted securities of the investor i over n years. As mentioned above, all values in Norwegian krone were inflation adjusted. Bank deposits include all forms of bank deposits, including transferable deposits, time deposits, and contractual deposits. The currency consists of notes and coins, and includes both domestic and foreign currencies. Regarding mutual funds, if investors own shares Norwegian mutual funds, the tax authority collects the value of mutual funds directly from the corresponding financial institutions; if investors own shares in foreign mutual funds, they are requested to self-report the value of funds for the tax purposes. Bonds and money market funds are treated in one category, and we only have the aggregated number of the two at the end of year. Finally, unlisted securities include shares in private firms and other securities that are not registered at the Norwegian Central Securities Depository.

[Insert Table 1 about here]

The summary statistics are presented in Panel A of Table 1. The data reveal significant variability between individuals in changes in stock allocation over one-, three-, and five-year windows, as indicated by a large standard deviation. On average, investors reduced their stock allocation in their financial assets by 2.12% one year after the crisis, the median reduction being 0.15%. Three years post-crisis, stock allocation appears to rebound slightly, with an average change of -0.07% and a median change of -0.003%, which are virtually zero. Similarly, five years after the crisis, the average change in stock

allocation is 0.17% with a median change of 0%. Regarding stock concentration, 14% of investors liquidated all their stocks one year after the crisis; this percentage increased to 21% and 27% did so three years and five years after the crisis, respectively. On average, the number of unique stocks in the investors' stock portfolio decreased across three time windows. Changes in the stock concentration index were negative one year after the crisis, but turned positive three and five years later, all with large standard deviations. For allocation in other financial assets, investors showed on average positive changes in bank deposits, bonds and money market funds, and unlisted securities, while there were almost no changes in mutual funds.[11]

2.3.3 Crisis experience and formative year experience

To measure crisis experience of investors, we start with the data at the transaction level. There are around 420,000 individuals who held stock at the end of year 2007, which is approximately 10% of the taxpayers in Norway. We identify all investors who held or traded stocks on the Oslo Stock Exchange from December 31, 2007 to December 31, 2008 and merge this sample with wealth and sociodemographic information. We require that all investors have data on wealth and some sociodemographic variables for at least one year before and five years after the crisis, that is, the period of 2007-2013. In addition, we remove the inactive investors with small holdings, those who did not trade at all in 2007 and 2008 and whose total portfolio holdings are worth less than 1000 Norwegian krone (NOK) or approximately 143 USD (1 NOK was traded at 0.143 USD on December 31, 2008) —- which is less than 1% of the sample. Our final sample consists of 344,209 investors.

To capture the personal crisis experience, we compute the stock portfolio loss using either the total loss or realized loss during the financial crisis for each individual investor. The total loss is the sum of the realized loss and the unrealized loss defined below.

$$
\begin{aligned}
&TotalLoss = RealizedLoss + UnrealizedLoss \\
&= \underbrace{\Big\{-\sum_{i=1}^{N_{sold}}\Big[\big(P^{s}_{i,2008} - P_{i,2008-}\big)\times Shares_i\Big]\Big\}}_{RealizedLoss} + \underbrace{\Big\{-\sum_{j=1}^{N_{held}}\Big[\big(P_{j,2008} - P_{j,2008-}\big)\times Shares_j\Big]\Big\}}_{UnrealizedLoss},
\end{aligned}
\tag{4}
$$

[11] At the end of 2008, the average investor had 461,303 NOK in cash deposits, 49,215 NOK in mutual funds, 19,877 NOK in bonds and money market funds, and 530,704 NOK in other unlisted securities.

where $P^s_{i,2008}$ is the selling price for stock i in 2008, $P_{i,2008-}$ is the purchasing price for stock i if it is bought in 2008 or its price at the end of 2007 if it's bought prior to 2008, $Shares_i$ is the number of shares of stock i sold in 2008, N_{sold} is the number of different stocks sold in 2008; $P_{j,2008}$ is the price for stock j at the end of 2008, $P_{j,2008-}$ is the purchasing price for stock j if it is bought in 2008 or its price at the end of 2007 if it's bought prior to 2008, $Shares_j$ is the number of shares of stock j held at the end of 2008, N_{held} is the number of different stocks held at the end of 2008. For every single stock held by an investor, we calculate the realized and unrealized losses and sum them up at the investor level (a negative number means profits) and then scale it with the investor's personal wealth at the end of 2007 (including stocks portfolio as well as bank deposits, mutual funds, bonds and money market funds, and other unlisted securities). We label this variable as *Total loss%*.

[Insert Table 2 about here]

Table 2 presents summary statistics for the characteristics of the investor stock portfolio. In our sample, investors held just over two stocks on average at the end of year 2008, with a median of only one stock (see *Number of stocks held*). The total value of all stocks held at the end of 2008 (*Value of stocks held*) by a median investor is approximately 376,000 NOK, while the average value is approximately 1.5 million NOK. About half of our sample conducted transactions during 2008. Among those who traded, the average number of transactions within the year is 3.7, with a median of two transactions (see *Number of transactions*). Regarding the total value of transactions, we calculate the sum of all transactions within the year for each individual by summing up the absolute value of each transaction. The average value of all transactions in a year, *Value of transactions*, is approximately 4.4 million NOK, while the median value is 56,000 NOK.

Next, we calculate the crisis experience, *Total loss* and *Realized loss*, as defined in Equation (4) above. For the full sample, the average *Total loss* is about 160,000 NOK, representing 12% of wealth (see *Total loss%*). In a subsample of investors who made at least one transaction in 2008, the average *Realized loss* is 224,000 NOK, with a large variation, ranging from a gain of about 2,200 NOK in the 10th percentile to a loss of 460,000 NOK in the 90th percentile.

We construct crisis experience measures, *Large loss dummy*, based on the variable

Total loss%.[12] Since most investors experienced realized and unrealized losses to some extent, we apply two different thresholds to identify a large loss. Specifically, *Large loss dummy* equals one if *Total loss%* lies above the 90th percentile ($Total\,loss\% > 33.1\%$) or the 75th percentile ($Total\,loss\% > 13.3\%$).[13] By design, the higher the threshold, the smaller the treated group and the larger the shock experienced by an investor. The summary statistics are shown in Table 1 (Panel B).

In addition to personal experiences during the financial crisis, we also consider formative year experiences, which can affect financial decisions Malmendier and Nagel (2011, 2016); Malmendier et al. (2020b,a). For each investor in our sample, we construct a variable, *formative year Experience*(FYE), representing the cumulative stock market returns when they experienced between the ages of 11 and 20, in the spirit of Malmendier and Nagel (2011).

Specifically, for each individual i, we calculate a weighted average of past stock market returns as follows:

$$FYE_i(\lambda) = \sum_{t=11}^{20} w_{i,t}(\lambda) R_t \tag{5}$$

where R_t is the stock market return in year t. The weight $w_{i,t}(\lambda)$ is defined as:

$$w_{i,t}(\lambda) = \frac{(t-10)^\lambda}{\sum_{t=11}^{20}(t-10)^\lambda} \tag{6}$$

We include market returns during the formative years (ages 10 to 20) when beliefs and preferences have been shaped. We follow Malmendier and Nagel (2011) by introducing a weighting function and a parameter λ that allow us to account for the possibility that recent experiences may have a different influence compared to a more distant past. When $\lambda = 0$, equal weights are assigned to each year from age 11 to 20, effectively taking the moving average of 10-year returns. A positive λ assigns higher

[12]We mainly consider total loss because an investor does not have to realize a loss to feel the pain – the losses on paper could well be enough. In a robustness check, we also test our main analysis using dummies derived from the alternative measure, *Realized loss%*, which only includes the realized losses for a sub-sample of investors who traded stocks during the year. The results are similar.

[13]In the robustness checks, we also apply the threshold at the median, creating *Large loss dummy* equals one if *Total loss%* lies above the 50th percentile ($Total\,loss\% > 4\%$). We find similar results in terms of signs and significance of all coefficients, except that the magnitude of the coefficients on *Large loss dummy* is smaller using a more lenient threshold.

weights to the more recent years. [14]

We observe a large variation the formative year returns.[15] Those born in the 1900s and the late 1950s experienced negative returns, while those born around 1920, 1950, 1960-70, and 1985 experienced largely positive returns. The patterns remain consistent regardless of whether the weighting parameter is set to $\lambda = 0$ or $\lambda = 1.5$. Not surprisingly, the formative year returns using $\lambda = 0$ is smoother than those using $\lambda = 1.5$. The summary statistics are shown in Panel B of Table 1. The average *FYE* is 4.6% and 4.8% for $\lambda = 0$ and $\lambda = 1.5$, respectively.

2.3.4 Wealth and sociodemographic characteristics

To control for investor characteristics, we include a number of control variables related to personal wealth and sociodemographic characteristics that can influence investor investment decision. All wealth-related control variables are measured at of the end of 2007, before the 2008 financial market crisis. All financial values in the raw data are in thousands of NOK and are inflation adjusted using CPI.

Personal wealth is measured as the natural logarithm of total personal wealth, encompassing financial capital (bank deposits, shares in stock funds, bonds and money market funds, stocks and other unlisted securities) and real capital (estimated market value of primary dwellings, secondary dwellings, and land and buildings used for business activity and other types of nonfinancial assets, e.g., forests, farms, machinery, inventories, etc.). *Annual salary* is the natural logarithm of the sum of employee income and net income from self-employment earned during the calendar year. *Personal liabilities* is the natural logarithm of outstanding personal liabilities owed to Norwegian and/or foreign creditors. *Dividend income* is the natural logarithm of share dividends received during the calendar year. *Social security recipient* is a dummy variable that equals one if the person received social security, and zero otherwise. *Ownership in private firms* is a dummy variable that equals one if the person owns shares in private firms, and zero otherwise. *Self-employed* is a dummy variable that equals one if the person is self-employed and zero

[14] For illustration of the weighting function, we include in the Online Appendix (Figure A.1) the weights $w_{i,t}$ as a function of the time t for two values of $\lambda = 0$ and $\lambda = 1.5$. Malmendier and Nagel (2011) use $\lambda = 1.5$ for their main analysis.

[15] In the Online Appendix, Figure A.2 depicts the formative year experience using the weighting parameters $\lambda = 0$ and $\lambda = 1.5$ for investors born between 1905 and 1990. We observe a large variation in the formative year returns.

otherwise.

In addition, we control for the age, gender, marital status, and education of the individual measured in 2008, which are used as a proxy for the risk propensity of individuals or financial sophistication in the prior literature. *Age* is measured as the natural logarithm of the age of an investor. *Married* is a dummy variable that equals one if a person is married or entered into a registered partnership, and zero otherwise. *College degree* equals one if a person's highest level of education is college / university or higher, and otherwise zero. *Econ/Business education* is a dummy variable that takes a value of one if an investor got a college/university degree in the field of business administration, economics, or finance, and otherwise zero.

The control variables are summarized in Table 1 Panel B. To mitigate the impact of potential outliers, we winsorized our variables at the 1% and 99% tails. On average, the investors in our sample have personal wealth of 1,938,893 NOK (approximately 277,000 USD in 2007), annual income of 415,184 NOK (59,312 USD), personal liabilities of 903,657 NOK (129,100 USD), and dividend income of 22,244 NOK (approximately 3,178 USD). Among all investors in our sample, 25% received some form of social security from the government, 11% had ownership in private firms, and 15% were self-employed. About 71% of the investors are male, with an average age of 52 years. Furthermore, 71% of the investors are married, about 43% of the investors have a college or university degree, and 10% have a degree in business-related studies.

3 Personal crisis experience and future stock allocation

In this section, we examine the relationship between investor crisis experience and their future stock allocation and future stock concentration.

3.1 Changes in stock allocation

To investigate the relationship between personal crisis experience and future stock allocation, we start with the following regression.

$$\Delta StockAllocation_i(n) = \alpha + \beta \text{Large loss dummy}_{i,2008} + \gamma X_{i,2007} + \theta P_{i,2008} + \epsilon_i \qquad (7)$$

where n represents the number of years (1, 3, or 5), and i denotes the individual investor. To explore different horizons, we measure the change in one, three and five years in future stock allocation (2009, 2011, and 2013 relative to the 2008 crisis year].[16] The independent variable *Large loss dummy*, defined in Subsection 2.3.3, is designed to capture personal crisis experience. Given that the vast majority of investors have experienced losses during the financial crisis, this variable differentiates between those who experienced large stock investment losses ($Large\ loss\ dummy = 1$) and those with smaller losses ($Large\ loss\ dummy = 0$). The vector $X_{i,2007}$ is a set of variables related to investor wealth measured before the crisis that can potentially affect individuals' investment decisions, including *Personal wealth*, *Annual salary*, *Personal liabilities*, *Dividend income*, *Social security recipient*, *Ownership in private firms*, and *Self-employed*. $P_{i,2008}$ is a set of personal characteristics measured in 2008, including *Male*, *Age*, *Married*, *College degree*, and *Econ/Business education*.[17]

We cluster standard errors along two dimensions: wealth and age cohorts. According to Abadie et al. (2023), there are two reasons for clustering the standard errors, the sampling process and the treatment assignment mechanism. Given that we have a full population of Norwegian investors, there is little concern over the sampling process — the correlations induced by sampling the outcome variable from a data-generating process with unobserved cluster-level components. However, we must consider the potential treatment effect in the sense that investors in the different wealth categories and age cohorts behave differently after having experienced a large loss in equity investment. Therefore, in addition to controlling for age and personal wealth together with other personal characteristics, we also cluster standard errors along the combination of wealth and age cohorts ($10 \times 10 = 100$) bins; where wealth and age are divided into ten groups each representing a decile).

[Insert Table 3 about here]

[16] Our method is similar to the collapsed difference-in-differences approach, where the data are collapsed into pre- and post-periods(see Bertrand et al. (2004)). To quantify the duration of the effect, we collapse the post-period over different horizons ranging from one to five years after the crisis. In the robustness check section and the online Appendix H, we present a detailed difference-in-differences analysis and observe similar patterns, confirming that personal crisis experience significantly influence investors' equity allocation decision.

[17] In Table A.1 in Online Appendix, we show that pre-crisis investor characteristics are correlated with the likelihood of experiencing large equity losses in 2008. Therefore, we control for this set of pre-crisis investor characteristics in all the regressions on stock allocation and concentration.

Table 3 shows the coefficients and standard errors of the regression estimations as specified in Equation (1). The first three columns report the results for regressions that include only control variables, without *Large loss dummy*. The next six columns introduce two different thresholds for defining what could be considered a large loss. Specifically, *Large loss dummy* equals one if *Total loss%* lies above the 90th and 75th percentiles, respectively. By construction, the higher the threshold, the smaller the treated group, and the larger the shock that an investor experienced.

The table reveals that all the coefficients for *Large loss dummy* are negative and significant, after controlling for a rich set of individual characteristics such as age, personal wealth, employment status, marital status and education. Importantly, the effect of crisis experience seems to persist, as the coefficients remain significant throughout the changes of one year, three years and five years and the magnitude of the coefficients also remained at about the same level. The coefficients in Columns 4-6 suggest that an experience of 33.1% loss (the 90th percentile) in the equity portfolio during 2008 is associated with a decline in the equity allocation by 3.4% from 2008 to 2009, by 4.6% from 2008 to 2011, and by 4.4% from 2008 to 2013. Similarly, the coefficients in Columns 7-9 show that investors who experienced a 13.3% loss (the 75th percentile) or greater in their equity portfolio during 2008 are likely to reduce the equity allocation by 2.7%, 2.5%, 2.6% for one-, three-, five-year change, respectively. Given that the average stock allocation 22.3% at the end of 2007 and 17.3% at the end of 2008, a reduction of 3.4% in one year is substantial — a 15% drop from the pre-crisis level or a 20% drop from the crisis year. Interestingly, the magnitude of the coefficients for *Large loss dummy* is much larger using a stricter threshold (columns 4-6) than those using a moderate threshold (last three columns). This perhaps suggests that the larger the loss an investor experienced, the smaller the proportion of financial assets that she would allocate to stocks in the future.

In sum, our results indicate that investors experiencing substantial losses in the financial crisis tend to reduce their allocation of financial assets to equities in the period up to the next five years. This impact of loss in a crisis is robust to controlling for a comprehensive set of personal characteristics. It is also robust to different thresholds of large losses in the financial crisis. These results show that the equity loss experience in a financial crisis has a prolonged impact on an investor's willingness to take equity market risk.

3.2 Changes in stock concentration

As we have shown that an average investor reduced the allocation to equities after a large loss in the 2008 crisis, it is natural to ask how the investor achieves the reduced equity exposure. We can think of at least three possibilities. First, investors could scale their investments more or less proportionally between stocks. In this case, we would expect their portfolio concentration to remain largely unchanged even though the allocation to equities was reduced. Second, investors could reduce their portfolio size while holding more diverse stocks, which would increase their portfolio diversification or lower their portfolio concentration. Third, investors could liquidate some stock holdings without reinvesting in other stocks. In such a case, we would expect their portfolio concentration to increase while the allocation to equities decreased.

As these three ways of reducing equity exposure have different implications on stock concentration, we will find out how investors achieve the reduced equity exposure after the financial crisis by studying changes in stock concentration. As described in Subsection 2.3.2, we capture changes in investors' stock concentration using three measures: stock liquidation, change in the number of stocks, and a concentration index.

First, we estimate the likelihood of future stock liquidation after having experienced a large loss during the crisis, using a Logit model.

$$Liquidation_i(n) = \alpha + \beta \text{Large loss dummy}_{i,2008} + \gamma X_{i,2007} + \theta P_{i,2008} + \epsilon_i \qquad (8)$$

where $n = 1, 3, 5$ represents the number of years after the financial crisis, and i denotes the individual investor. The dependent variable, $Liquidation(n)$, is a dummy variable that equals one if the investor liquidates all stocks in n year after the crisis and zero otherwise. This analysis includes all investors who held stocks at the end of 2008. The independent variable, *Large loss dummy*, defined as in Subsection 2.3.3, captures the personal crisis experience. In addition to all control variables included in Table 3, we also control for the number and the value of stocks held at the end of 2007 (logarithm). Unlike our prior regressions that measure relative change in stock allocation, $\Delta StockAllocation$, (where information from previous years is already captured), here we estimate the likelihood of complete liquidation in a particular year and thus explicitly control for initial portfolio characteristics.

Next, we examine changes in stock concentration among investors who remain in

the stock market (i.e., those who do not completely liquidate their stock holdings). We first measure changes in the number of stocks and estimate the following equation:

$$\Delta Numberof Stocks_i(n) = \alpha + \beta \text{Large loss dummy}_{i,2008} + \gamma X_{i,2007} + \theta P_{i,2008} + \epsilon_i \quad (9)$$

where $n = 1$, 3, or 5 represents the number of years after the crisis, and i denotes investor. The sample is restricted to investors who continue to hold at least one stock at the end of year 2008+n. The dependent variable $\Delta Numberof Stocks_i(n)$ is defined in Subsection 2.3.2. The independent variable, *Large loss dummy*, is defined as in Section 2.3.3. We use the same set of control variables, $X_{i,2007}$ and $P_{i,2008}$, as in Table 3, which includes *Age*, *Personal wealth*, *Annual salary*, *Personal liabilities*, *Dividend income*, *Social security recipient*, *Ownership in private firms*, *Self-employed*, *Male*, *Married*, *College degree*, and *Econ/Business education*.

In addition to changes in the number of stocks, we measure the change in the stock concentration index and run the following estimation:

$$\Delta StockConcentration_i(n) = \alpha + \beta \text{Large loss dummy}_{i,2008} + \gamma X_{i,2007} + \theta P_{i,2008} + \epsilon_i \quad (10)$$

This estimation is almost the same as the one above except that the dependent variable is replaced with $\Delta StocksConcentration = HHI_{2008+n} - HHI_{2008}$ that is defined in Section 2.3.2. As with the number of stocks analysis, this estimation is restricted to investors who remain in the stock market, ensuring that changes in concentration reflect portfolio rebalancing decisions rather than the mechanical effect of complete liquidation. The distinction between the liquidation analysis (Equation 8) and the concentration analyses (Equations 9 and 10) is important: the former captures the extensive margin decision of whether to exit the stock market entirely, while the latter capture the intensive margin decisions of how to rebalance the portfolio conditional on remaining invested.

The results are tabulated in Table 4. Panel A shows the probability of stock liquidation from the Logit model as specified in Equation (8). Panel B summarizes the results for changes in the number of stocks and stock concentration index as specified in Equation (9) and Equation (10). Please note that Panel A includes the full sample of investors as in Table 3 and Panel B focuses on a subsample of investors who did not completely liquidated their stock holdings. For the sake of space, we only tabulate the coefficients and standard errors of the main variables *Large loss dummy*. All the control variables described above are included in the regressions and are not tabulated in the

table. In addition, all standard errors are clustered along the combination of wealth and age cohorts bins as explained in Section 3.

[Insert Table 4 about here]

As shown in panel A of Table 4, all coefficients for *Large loss dummy* are positive and significant, after controlling for the number and the value of stocks held in 2007 as well as a set of individual characteristics such as age, personal wealth, employment status, marital status and education. In addition, we calculate the marginal effect of *Large loss dummy* by taking the difference between the probability of stock liquidation given *Large loss dummy*=1 and *Large loss dummy*=0 while holding all the control variables at their mean. The marginal effects in the first three columns and the last row of Panel A suggest that an experience of 33.1% loss (the 90th percentile) or more in the equity portfolio during 2008 is associated with a 3.3% increase in the likelihood of liquidating all of their stock holdings in 2009, 8.1% in 2011, and 10.6% in 2013. The marginal effects in the last three columns show a similar story for investors who experienced 13.3% (the 75th or more loss in their equity portfolio during 2008, the likelihood of liquidating all their stock holdings increases by 2.2% in 2009, 5.6% in 2011, and 7.8% in 2013.

Panel B examines portfolio concentration changes conditional on remaining in the stock market. Importantly, this analysis addresses a different question than Panel A: while Panel A captures whether investors exit the market entirely (the extensive margin), Panel B examines how continuing investors adjust their portfolio composition (the intensive margin). The sample in Panel B excludes all investors who completely liquidated their stock holdings, ensuring that our concentration measures reflect deliberate portfolio rebalancing decisions rather than the mechanical effect of selling all stocks.

The upper part of Panel B analyzes changes in the number of stocks following the personal crisis experience as specified in Equation (9). The coefficients *Large loss dummy* are all negative and statistically significant at the 1% level. Interpreting these coefficients, which represent the difference between the natural logarithm of the stock count in the portfolio of two years, a coefficient of -0.11 would mean that an investor who held 10 stocks in 2008 would only hold approximately 9 stocks in 2013; similarly an investors who held 2 stocks in 2008 would hold about 1.8 stocks in 2013. The negative and significant coefficients suggest that investors have experienced a large loss during the financial crisis,

reducing their stock count for one-year, three-year, and five-year horizons. Furthermore, when conducting the same analysis using the entire sample, we observe similar patterns that *Large loss dummy* is consistently associated with a decrease in the number of stocks for up to five years after the crisis (results not tabulated).

The bottom part of Table 4 Panel B shows the results on future stock concentration as specified in Equation (10). The dependent variable, *n-year change in stock concentration* is based on *HHI* and a positive change in *HHI* means an increase in stock concentration. The results show that all the coefficients on *Large loss dummy* are positive and significant, controlling for the set of individual characteristics as before. The crisis experience seems to have a lasting effect, as the coefficients remain significant across one-year, three-year, and five-year changes and that the magnitude of coefficients gets larger over the time. The coefficients in the first three columns indicate that after an equity loss of 33.1% (the 90th percentile) or greater, the stock concentration (*HHI*) of investors' equity portfolio increases on average by 1.6% from 2008 to 2009, 4.7% to 2011, and 5.9% to 2013. Similarly, the coefficients in the last three columns suggest that an experience of 13.3% or greater loss (the 75th percentile) in the equity portfolio during 2008 is associated with an increase in the *HHI* by 1.2% from 2008 to 2009, 4.5% to 2011 and 5.4% to 2013. Furthermore, the magnitudes of the coefficients on *Large loss dummy* are comparable across different definitions of the large loss.

In summary, our results suggest that investors experiencing a large loss in the financial crisis tend to increase stock concentration in subsequent years. This impact of crisis loss is robust to different thresholds of large losses and to controlling for personal characteristics. This finding is consistent with the conjecture that investors reduce the allocation to equities by liquidating some stock holdings without reinvesting in other stocks. Whereas investors intend to reduce their risk exposure to equities after experiencing a large equity loss, one unintended consequence is the increased idiosyncratic risk of their equity portfolios, which may work against their desire to take less stock market risk.

3.3 Analysis using the siblings subsample

In the previous section, our baseline models included a comprehensive set of control variables that capture factors that could influence investment decisions of individuals. Nonetheless, there remains a possibility that unobserved variables

contribute to the variation in these decisions. To mitigate this concern, we conduct an additional analysis using a subsample of siblings, as studies suggest that genetic differences can account for some of individual variations in risk-taking and financial-decision makings (see, for example, Cesarini et al. (2009) and Cesarini et al. (2010)).

Employing matched sibling pairs as an empirical strategy offers several advantages. Siblings share a considerable portion of their genetic makeup and often grow up in similar environments, which helps control for unobserved familial factors that could influence financial behavior. By focusing on sibling pairs, we can more effectively isolate the impact of unique individual experiences and external economic conditions on investment decisions, thus enhancing the robustness of our analysis. This approach is particularly valuable for addressing the potential biases introduced by unobserved heterogeneity that may confound the relationship we are investigating.

3.3.1 Sibling sample

We start with the full population registry data from Statistics Norway and exclude individuals who were under 18 years old in 2007. Siblings are identified using parental ID, ensuring that they share the same mother and father. To be included in the sample, there must be at least two siblings within a sibling pair that held stocks before the crisis, with at least one sibling experiencing a large loss while the other(s) did not. We also require all individuals to have records in the tax data and demographics data from 2007 to 2013. This results in a sample size of 10,801 (using the 90th percentile) or 22,311 investors (using the 75th percentile) for the thresholds of *Total loss %*.

[Insert Table 5 about here]

Table 5 Panel A presents descriptive statistics on the subsample of matched sibling pairs that are further divided into groups—those who experienced a large loss during the crisis and those who did not. To assess differences between these two groups of individuals, we perform the two-sample t-test for means and Mann-Whitney U rank-sum test for medians, with the p-values displayed in the last two columns of the table. The numbers indicate significant differences between those who have experienced a large equity loss during the crisis and those who did not across various dimensions, even among

siblings. This highlights the importance of considering sibling pairs to account for both observed and unobserved factors influencing investment behavior.

3.3.2 Change in stock allocation after the crisis

We replicate the baseline results for stock allocation and concentration after the crisis, as shown in Table 3 and 4, by incorporating siblings fixed effects. In addition, we create ten dummies based on ten deciles of pre-crisis stock allocation to ensure comparisons are made among individuals with similar stock allocation before the crisis. Note that fewer observations are available in the analysis of (8) as regression estimations require variations within a sibling pair in the dependent variable, *Stock liquidation dummy*.

The results are summarized in Panel B of Table 5. The top panel shows the results for the change in stock allocation. The coefficients for *Large loss dummy* are all negative and significant for the one-, three-, and five-year change in stock allocation. This is in line with our baseline results that investors tend to reduce their allocation of financial assets to equities after experiencing a large loss in the financial crisis. The next panel presents the results for Logit regressions on the likelihood of stock liquidation. The positive and significant coefficients suggest a higher likelihood of future stock liquidation after having experienced a large loss during the crisis, consistent with the results for the full sample.

For the subsample of siblings who have not fully liquidated their stock holdings, the negative and significant coefficients on the change in number of stocks confirm the baseline results that the number of stocks reduces after investors have experienced a large loss during the financial crisis. Finally, the bottom panel shows the results for the change in stock concentration. The coefficients for *Large loss dummy* are all positive and significant, confirming our previous finding that investors experiencing a large loss in the financial crisis tend to increase stock concentration in the years following.

To sum up, the baseline results are robust and strong in the siblings analysis. Within the matched sibling pairs, those who experienced a large equity investment loss allocate fewer financial assets to stocks and are more likely to liquidate stocks in the years following the crisis. If they maintain a stock portfolio after the financial shock, they tend to increase the portfolio concentration by investing in a smaller number of stocks. These results are consistent with our previous findings that a personal loss experience in a

financial crisis has a prolonged impact on an investor's willingness to take equity market risk.

3.4 Changes in other financial asset

So far, we have demonstrated that investors who suffered extreme stock market losses during the financial crisis tend to allocate less wealth to stocks, hold more concentrated stock portfolios, and show a stronger tendency to exit the stock market. The next question is whether they shift their allocation in other financial assets?

We examine the effect of personal crisis on future changes in the allocation of other financial assets by estimating the following regressions.

$$\Delta Y_i(n) = \alpha + \beta \text{Large loss dummy}_{i,2008} + \gamma X_{i,2007} + \theta P_{i,2008} + \epsilon_i \tag{11}$$

where (n = 1, 3, or 5) is the number of years after the financial crisis, $\Delta Y_i(n)$ is the n-year change in either cash deposit, mutual funds, bonds and money market funds, or other unlisted securities defined in Subsection 2.3.2, and i denotes investor. We use the same set of controls, $X_{i,2007}$ and $P_{i,2008}$, as in our baseline equation 1.

As mentioned in Section 2, we have annual data on other financial assets, including cash deposit, mutual funds, bonds and money market funds, and other unlisted securities. However, we do not have any information on the transactions and holdings of these items. Instead, we only have the end-of-year value that is collected from the Norwegian tax authority. Therefore, we can only have coarse measures of changes in the allocation of these asset classes based on the yearly end-of-year values. Unlike our measure for stock allocation, we cannot distinguish between active changes in the allocation and passive changes resulting from changes in the market value of these assets. Thus, we need to interpret the numbers with caution. Nevertheless, it is still valuable to observe changes in other financial assets within an investor's portfolio. Table 6 presents the coefficients and standard errors for the main variable *Large loss dummy* is reported in. All other control variables are included in the regressions and are not tabulated in the table.

[Insert Table 6 about here]

Changes in bank deposit. As shown in the top panel of Table 6, coefficients for *Large loss dummy* are positive and significant for one-year and three-year changes.

For five changes in cash deposit, the coefficients for *Large loss dummy* is only significant when using the 75th percentile threshold for large loss. The positive and significant coefficients suggest that having experienced a large loss in stock investment during the crisis is associated with an increase in bank deposit between 1% to 2% over one- and three-year periods.

Changes in mutual funds. The second panel shows the results for the change in equity fund allocation. The coefficients are all negative and significant, indicating that investors who experienced a large loss in their equity portfolio during 2008 are likely to reduce the allocation to mutual funds. When using the 90th percentile threshold for the large loss dummy, the reductions in mutual funds allocation are 0.81% in one year, 0.97% in three years, and 2.5% in five years after the crisis. These changes are substantial, considering the pre-crisis allocation in mutual funds was about 10%.

Changes in bonds and money market funds. The third panel of Table 6 presents regression estimation results for changes in bonds and money market funds. The coefficients for *Large loss dummy* are positive and significant over 3- and 5-year horizons, suggesting that investors tend to increase their holdings in bonds and money market funds after having experienced a large loss in the 2008 crisis. The increases in the allocation of wealth to bonds and money market funds are approximately 0.1% in three years and 0.3% in five years. Given the pre-crisis allocation to bonds and money market funds was about 1.5%, these increases are considerable.

Changes in unlisted securities. The bottom panel reports the result of changes in unlisted securities. The coefficients are mostly insignificant, except for the three-year change in unlisted securities using the 90th percentile as the threshold for the large equity loss. Therefore, we do not observe significant change in the allocation to unlisted securities.

In short, the results seem to indicate that investors who have experienced a large loss in the financial crisis tend to increase their cash holdings and holdings in bonds and money market funds while reducing their allocation to mutual funds. These results are consistent with our previous findings that a personal loss experience in a financial crisis has a prolonged impact on an investor's willingness to take equity market risk. However, it is important to remember that due to the data limitation, we cannot distinguish between active changes in the allocation and passive changes.

3.5 Potential financial impact

We have established that investors who experience a large loss during the financial crisis tend to reduce their allocation of financial assets to equities and increase stock concentration in subsequent years. Many of these investors have done so by liquidating some or even all of their stock holdings without reinvesting in other stocks. A natural follow-up question would be to quantify the financial impact of such investment behaviors.

We try to calibrate the potential welfare consequences by comparing investors who experienced a large loss in the financial crisis versus those who experienced a smaller loss. Specifically, for each individual, we calculate *Difference* as the difference between two values:

- **Actual (active) investing** The end-of-year portfolio value plus income from all stock sales minus expenses from all stock purchases during the period;
- **Hypothetical (passive/buy-and-hold) investing** A benchmark that assumes the investor would hold the stocks in her portfolio at the end of 2008 for some year.

A positive *Difference* indicates that the investor was better off changing the stock allocation from her portfolio in 2008 whereas a negative *Difference* indicates a loss from the change in her stock allocation. We then sort investors into two groups: those who have experienced large stock investment losses in 2008 ($Large\,loss\,dummy = 1$) and the rest who experienced smaller losses ($Large\,loss\,dummy = 0$) .

[Insert Table 7 about here]

The results are shown in Table 7. By construction, the number of observations in each group varies depending on the threshold used for *Large loss dummy*. On average and by median, *Difference* is negative for both groups. However, the financial impact is more severe for the group that experienced large losses during the 2008 crisis. Specifically, for the group of investors whose losses exceeded the 90th percentile, their average potential welfare loss over a five-year period is 149,496 NOK and the median loss is 50,007 NOK. In contrast, for the group of investors who experienced a smaller loss during the financial crisis, the average potential welfare loss is 11,984 NOK and the median is 202 NOK. In addition, we conduct a two-sample t-test to compare the means of two groups and a

nonparametric two-sample test on the equality of medians. Both tests indicate that the difference between the two groups is statistically significant. This pattern holds across different time windows and different thresholds for the *Large loss dummy*.

In summary, we find that investors who experienced large losses during the financial crisis tend to reduce their stock allocation and increase stock concentration by liquidating some stocks without reinvesting in others. Such investment behaviors have negative and sizeable financial consequences.

3.6 Expectation of stock market returns

How should we understand the impact of investment loss during the financial crisis on an investor's equity allocation decision? To shed light on this question, we take advantage of insights from a standard model of risky asset allocation (Campbell and Viceira (2002), Chapter 6). In this model, an investor's risky asset allocation depends on three key parameters: expected excess return on the risky asset, volatility risk of the risky asset, and (relative) risk aversion.

Volatility risk. The volatility risk in the stock market is observable. For instance, the Implied Volatility Index for the Norwegian equity market (NOVIX) spiked in October 2008, reaching over 60% and then trending downward in the following five years (see Figure 1 Panel B). Realized volatility shares a similar time trend during this period. Hence, the decreased stock market allocation and increased tendency to exit the stock market after the financial crisis cannot be attributed to increased stock market risk.

Risk aversion. Risk aversion of individual investors is not directly observable. According to Campbell and Viceira (2002), we may assume a constant relative reversion for each individual investor, consistent with the observation that risk premia tend to be stable over the long term in a growing economy.[18]

Expected excess return. Of the three key parameters in Campbell and Viceira (2002), the only one that remains to be considered is changes in investors' expected

[18]Guiso et al. (2018) use an investor portfolio and survey data from an Italian bank to study how risk aversion fluctuates around the 2008 financial crisis. They report a substantial increase in investors' absolute risk aversion after the crisis, which is primarily driven by changes in perceived probabilities of return distribution and emotion-based changes of the utility function. Our data do not allow us to dynamically estimate investors' risk aversion coefficients. Instead, we draw on the literature on expectation formation to study subjective expectation of returns.

stock market returns following a financial crisis. Previous studies have examined the formation of subjective expectations in macroeconomics and finance based on surveys of economic agents (see Bordalo et al. (2022) for a recent survey). In financial markets, a key insight from the literature is the distinction between expected returns and expectations of returns, which points to departures from "rational" expectations.[19] For example, Greenwood and Shleifer (2014) study six data sources for investor forecasts of stock market returns; they find that expectations of future stock returns tend to be overly pessimistic after market crashes and overly optimistic after booms.

First, we estimate the dividend yield-implied expected returns over the one- to five-year windows using the dividend yield model $R^e_{t:t+k} = \alpha + \beta \times D_t/P_t + \epsilon_{t+k}$ as in Cochrane (2011). The dependent variable $R^e_{t:t+k}$ is the stock market return (obtained from Datastream OSE Industry total return index) less the 1-month government bond rate (based on government securities and NIBOR). The variable D_t/P_t is the dividend yield. Data are annual for the period 1970–2008. Following Cochrane (2011), the regression t-statistic uses the Hansen and Hodrick (1980) correction for the two to five year horizons. $\sigma[E_t(R_e)]$ represents the standard deviation of the fitted value, $\sigma(\hat{\beta}) \times D_t/P_t$. Risk-free rate for 1-, 3-, 5-year horizons are the government bond yield for the maturity of 1 year, 3 years, and 5 years, respectively. The risk-free rate for 2- and 4-year horizons are extrapolated based on the 1-, 3-, and 5- year rates.

Second, we try to measure the expectations of future stock returns as the literature shows that the subjective returns could depart from rational expectations. Although we do not have investor survey data in Norway on their stock market return expectations, it is reasonable to believe that individual investors across countries share a similar mechanism in forming their subjective expectations. Therefore, the highly positive correlation between expectations of returns and past stock returns may hold not only for US investors but also for Norwegian investors. Thus, we impute the expectations of returns using the models from Greenwood and Shleifer (2014). In their model, survey expectations of future stock market returns are regressed on past stock market returns and price-dividend ratio. We replace the stock market returns with the personal past 12-month stock returns and estimate the expected returns for each individual investor using:

[19] We need to interpret "rational" with a caution as it's not necessarily irrational for past experiences, particularly large shocks from crises, to alter preferences, leading to high discount rates, increased risk aversion, negative expectations, and even disutility for equity ownership. Thus, while realized returns were lower than counterfactual, investors may still be maximizing their utility by acting in such a way.

Model 1: $Exp_t = 5.789 + 3.133 R_{t-12:t-1}$

Model 2: $Exp_t = -7.979 + 3.354 R_{t-12:t-1} + 3.404 log(P_t/D_t) + \mu$

where $R_{t-12:t-1}$ is the past 12-month stock returns calculated for each individual and $log(P_t/D_t)$ is the the price-dividend ratio for the stock market. The estimated returns are truncated at -100% and 100%. The estimates are then used to understand their stock allocation behavior.

[Insert Table 8 about here]

In Table 8, we estimate the dividend yield-implied expected stock market returns and expectations of future stock returns at the end of 2008. Panel A shows that the forecast future returns, either as excess returns or gross returns, are extremely large and positive. In contrast, Panel B indicates largely negative expectations of future stock returns for 2009, differing from the dividend yield-implied expected stock market returns. For many individual investors who experienced large losses in the 2008 crisis, their expectations of stock market return in excess of the risk free are especially negative. Interestingly, because the aggregate dividend yield in the Norwegian stock market spiked toward the end of 2008, the dividend yield-implied expected stock market return is extraordinarily high: the average excess return, estimated from the predictive regressions as in Cochrane (2011), approximates 140% per annum in the five years following the crisis.

These findings suggest that the negative expectation of stock market risk premiums for those individual investors experiencing large losses during the financial crisis might explain their reduced stock market allocations and increased tendency to exit the stock market after the crisis when increased stock market allocations are most attractive.

4 formative year experience

Prior literature suggest that experiences may influence financial decision making through the formation of beliefs and preferences (Malmendier and Nagel, 2011, 2016; Malmendier et al., 2020b,a). Building on insights from the experience effect literature, this section explores the impact of the crisis experience along the formative year experience.

Following Malmendier and Nagel (2011), we construct a variable, *formative year Experience (FYE)*, that represents the cumulative stock market returns when an individual investor was 11 to 20 years old. The detailed definition of this variable is provided in Section 2.3. To examine the effect of personal crisis experience together with formative year experience on stock allocation decisions, we add *formative year experience* (FYE) and its interaction term with *Large loss dummy* to our baseline equations (7), (9), and (10). Specifically, we estimate the following regression:

$$\begin{aligned}\Delta Y_i(n) = \alpha + \beta &\text{Large loss dummy}_{i,2008} \times \text{FYE}_{i,2008} \\ &+ \gamma\text{Large loss dummy}_{i,2008} + \delta\text{FYE}_{i,2008} + \theta X_{i,2007} + \theta P_{i,2008} + \epsilon_i\end{aligned} \tag{12}$$

where Y is either *StockAllocation*, *Number of stock*, or *StockConcentration*; n = 1, 3, or 5 for the changes over one-, three-, or five-year window, and i denotes investor. The independent variable *Large loss dummy*, defined in Subsection 2.3.3, captures personal crisis experience. We include the same set of controls, $X_{i,2007}$ and $P_{i,2008}$, as before, including *Personal wealth*, *Annual salary*, *Personal liabilities*, *Dividend income*, *Social security recipient*, *Ownership in private firms*, *Self-employed*, *Age*, *Male*, *Married*, *College degree*, and *Econ/Business education*.

To examine the likelihood of future stock liquidation after having experienced a large loss during the crisis, we use a Logit model and add *formative year experience* (FYE) and its interaction term with *Large loss dummy* to Equation (8)

$$\begin{aligned}\textit{Liquidation}_i(n) = \alpha + \beta &\text{Large loss dummy}_{i,2008} \times \text{FYE}_{i,2008} \\ &+ \gamma\text{Large loss dummy}_{i,2008} + \delta\text{FYE}_{i,2008} + \theta X_{i,2007} + \theta P_{i,2008} + \epsilon_i\end{aligned} \tag{13}$$

where n = 1, 3, 5 represents the number of years after the financial crisis, and i denotes the individual investor. The dependent variable, *Liquidation*(n), is a dummy variable that equals one if the investor liquidates all the stocks in n year after the crisis and zero otherwise. We include all control variables as in Equation (8) and Table 4, Panel A.

[Insert Table 9 about here]

The regression results are reported in Table 9 with four panels summarizing the results for changes in stock allocation, likelihood of stock liquidation, changes in number of stocks and changes in stock concentration, respectively. We only tabulate the coefficients and standard errors of the main variables. All other control variables are included in

the regression and are not tabulated. In this table, we define *FYE* using $\lambda = 1.5$ as in Malmendier and Nagel (2011), meaning that higher weights are assigned to more recent years.[20]

First, the coefficients of *Large loss dummy* are similar to those of Table 3 and Table 4. They are all negative and significant in the top panel where the dependent variables are changes in stock allocation. The magnitudes of these coefficients are also comparable with those in Table 3. For example, the coefficients in the left columns suggest that an experience of 33.1% loss (the 90th percentile) in the equity portfolio during 2008 is associated with a decline in the equity allocation of about 4% from 2008 to 2009, by 5.1% from 2008 to 2011, and by 4.7% from 2008 to 2013. In comparison, the corresponding numbers 3.4%, 4.6%, and 4.4% for the baseline regressions without formative year experience. Regarding the likelihood of stock liquidation, we also find similar results: investors who experienced large losses during the financial crisis, the likelihood of liquidating all their stock holdings increases from one year to five years after the crisis. Furthermore, condition on staying in the stock market, investors who experienced a large loss during the financial crisis tend to reduce the number of stocks in their portfolio and increase their stock portfolio concentration over one-year, three-year, and five-year horizons. These findings indicate that our previous findings that individual investors experiencing a large loss in the financial crisis tend to reduce their stock allocation and increase their stock concentration are robust to controlling for formative year experience.

Second, *formative year experience* also appears to influence investment decisions alongside personal crisis experience. For example, in the upper panel of Table 9, the coefficients for *formative year experience* are positive and significant for 3-year and 5-year changes in stock allocation, suggesting an increase in stock allocation for those who experienced high formative year returns. The interaction term's coefficients are positive and significant for all three time horizons, indicating that among the investors who experienced large losses during the financial crisis, those who grew up with higher formative year returns are associated with a larger increase in stock allocation for up to five years after the crisis. Regarding the likelihood of stock liquidation, the coefficients for the interaction term are not statistically significant. The third panel shows results

[20]Malmendier and Nagel (2011) use $\lambda = 1.5$ for their main analysis. We also report results using $\lambda = 0$ to construct *FYE* in the online Appendix, Table A.5. The signs and magnitudes of all main variables are very similar to those using $\lambda = 1.5$.

for changes in the number of stocks after the crisis, where the interaction term's coefficients are mostly negative and significant, and the coefficients for *formative year experience* are only significant when *Large loss dummy* is defined using the threshold of the 90th percentile. The bottom panel presents regression results for changes in the stock concentration index, where five out of six coefficients for the interaction term are positive and significant and only half of the coefficients for *formative year experience* are significant. These results suggest that among investors who experienced a large loss in the 2008 crisis, their formative year returns are somewhat associated with a reduction in the number of stocks and an increase in stock concentration.

These findings indicate that the personal crisis experience and the stock market performance during investors' formative years together have a lasting impact on their equity allocation decision. First, investors experiencing a large loss in the financial crisis tend to reduce their stock allocation and increase their stock concentration, whereas investors who grew up with a bullish stock market return tend to increase their stock allocation. Second, among those who experienced large losses during the crisis, a high formative-year stock market return is associated with a larger increase in stock allocation and stock concentration. This result supports Malmendier and Nagel (2011)'s finding on the importance of investor experiences to financial risk taking.

In summary, we show that investors' personal crisis experience has a significant impact on their equity allocation decision in addition to past formative year experience. We also show that formative year experiences interact with crisis experiences: investors with positive formative stock market experiences partially offset their crisis-driven risk reduction, while those whose crisis experience resonates with negative formative experiences show intensified risk aversion. Our findings demonstrate that both direct personal losses and early-life market experiences jointly shape investor beliefs and drive persistent heterogeneity in risk-taking behavior long after financial crises.

5 Robustness checks

In this section, we check the robustness of our results to several variations in methodology, sample, and measurements. All the additional analyses are presented in detail in the Online Appendix.

Total loss% as an alternative measure. In our main analysis, we mainly rely on the

dummy variable *Large loss dummy*, based on the variable *Total loss%*, applying different thresholds to identify a large loss. The rationale is to focus on larger losses, as most investors experienced losses during the crisis year. However, instead of dummy variables, we can also use continuous variable *Total loss%* directly in our regressions. Thus, we re-run all our baseline regressions Equation (7)-(10) by replacing *Large loss dummy* with *Total loss%*. The results are presented in Table A.2 in the Online Appendix. We find similar results in terms of signs and significance of all coefficients. These results suggest that an increase in *Total loss%* is associated with a decrease in investor stock allocation, a higher likelihood of liquidation of their stock holdings, a decrease in the number of stocks and an increase in the stock concentration index in subsequent years. It confirms our previous finding that the crisis experience has impact on future stock allocation and stock concentration for at least up to five years.

Realized loss% as an alternative measure. In the previous analysis, our main explanatory variable, *Large loss dummy*, is based on the total loss during the 2008 crisis which is the sum of the paper loss and the realized losses. However, one could argue that the paper loss and the realized loss are different and that investors feel more direct pain from the realized losses. Therefore, we replicate our baseline regressions using the realized loss to construct the *Large loss dummy*. Please note that this measure is only available for a subsample of investors who made at least one transaction in 2008, meaning that investors who only experienced paper losses are not included in this analysis. The summary statistics of the realized loss are shown in Table 2. On average, *Realized loss* counts for 15.9% of financial asset. Similarly to the analysis performed previously, we apply two different thresholds to define large loss. Under the new definition, *Large loss dummy* equals one if *Realized loss%* lies above the 90th percentile (above 45.8%) and the 75th percentile (above 19.7%), respectively. The regression results are shown in Table A.2. Almost all coefficients are significant and exhibit the same signs as those based on *Total loss%* for the full sample; the magnitudes of the coefficients are also similar. All of these findings indicate that investors who have experienced a large realized loss in the financial crisis are associated with a decrease in the stock allocation and the number of stocks in the portfolio and an increase in stock concentration in the years afterward. The impacts of the crisis experience are robust to the alternative measure using realized losses and the subsample of investors who traded during the crisis.

A different threshold to define large losses. In our main text, all the regression

results are shown for the the *Large loss dummy* using the 90th and 75th percentiles of *Total loss%* to identify a substantial loss. In this section, we apply the median threshold as a robustness check, creating *Large loss dummy* equals one if *Total loss%* lies above the 50th percentile ($Total\,loss\% > 4\%$) and zero otherwise. As mentioned in the main text, the higher the threshold, the smaller the treated group and the larger the shock experienced by an investor. Thus, lowering the threshold to the median means that more investors are in the treated group, with on average fewer extreme losses within the treated group. We re-run all our baseline regressions, applying the median *Total loss%* as the threshold for the *Large loss dummy* and present the results in Table A.3. All the coefficients for *Large loss dummy* are statistically significant and exhibit the same signs as in the baseline regressions. The magnitudes of the coefficients are slightly smaller than those using the stricter thresholds. This perhaps suggests that the larger the loss an investor experienced, the stronger the effect on investors' future investment decisions. In summary, our findings that investors intend to reduce their risk exposure to equities and increase the concentration of stocks after experiencing a large loss of equity are robust to a more lenient threshold (the median *Total loss%*) for the *Large loss dummy*.

Different weighting parameter for formative year experience (FYE). In the main text, we show results using the weighting parameter $\lambda = 1.5$ to construct investors' formative year experience. This parameter assigns a higher weight to the more recent years and a lower weight to a distant past. Alternatively, we can use $\lambda = 0$, which assigns equal weights to all years, effectively taking the simple moving average of 10-year returns during the formative years of an investor. The weighting function and the formative year experience are illustrated in Figures A.1 and A2. We replicate all the regressions in Table 9 using $\lambda = 0$ to construct *FYE*. The results are summarized in Table A.4. The signs and magnitudes of the coefficients for *Large loss dummy* are very similar to those using $\lambda = 1.5$. In addition to personal crisis period, *formative year experience* also appears to influence investment decisions. Although some signs of coefficients for *FYE* differ from those using $\lambda = 1.5$, they do not affect the signs and significance of coefficients for *Large loss dummy*. The signs and significance of the interaction term are also similar to those using $\lambda = 1.5$. Thus, our previous conclusions that the personal crisis experience and the stock market performance during investors' formative years together have a lasting impact on their equity allocation decision are robust to controlling for formative year experience, whether using $\lambda = 0$ or $\lambda = 1.5$ as the weighting parameter.

Including pre-crisis stock portfolio returns. In addition to the formative year

experience of stock market returns, personal experiences with past stock portfolios can also shape investors' future investment behaviors. For instance, Barber and Odean (2001) find that past trading success leads to overconfidence, particularly among men, which influences their future trading behavior. Malmendier and Nagel (2011) that study the formative year experience also shows that more recent return experiences have stronger effects. As a robustness check, we construct a variable *Pre-crisis return*, which represents the value-weighted total stock return (both realized and unrealized) for each individual investor based on their tradings and holdings of the stock portfolio in 2007. We include *Pre-crisis return* in the baseline regressions and summarize the results in Table A.5. All coefficients *Large loss dummy* are significant and have the same signs as those based on the main text. The magnitudes of all coefficients are virtually the same after controlling for the pre-crisis stock portfolio returns at the individual investor level. Coefficients for *Pre-crisis return* are largely significant as well. They are negatively associated with future changes in stock allocation, the likelihood of stock liquidation, and changes in stock concentration, and are positively correlated with changes in the number of stocks. These results confirm our previous findings that the personal crisis experience has a lasting impact on investors' equity allocation decision, even after accounting for personal pre-crisis stock portfolio returns.

Full sample analysis for changes in the number of stocks. To capture changes in investors' stock concentration, we analyzed in the main text changes in the number of stocks and stock concentration index for a subsample of investors who did not completely liquidated their stock holdings. This focus was necessary because we cannot calculate the stock concentration index for non-stock holders. However, for changes in the number of stocks, which is the difference between the number of unique stocks held at the end of this period and that of the previous period, we do not need to limit our analysis to this subsample. For non-stockholders, the number of stocks is simply zero. Thus, we re-run the baseline regression specified in Equation (9) for the full sample. The results are presented in Table A.6. The coefficients *Large loss dummy* are all negative and statistically significant at 1% level, consistent with the results in Table 4. With respect to the magnitude of the coefficients, they are slightly more negative than those for the subsample of investors who remained in the stock market after the crisis. This is not surprising, as those who liquidated all the stocks are now included in the sample, resulting in more drastic drops in the number of stocks from a positive number to zero. In short, our previous finding that personal crisis experience is associated with changes

in the number of stocks in investors' portfolio is robust when applied to the full sample.

Difference-in-differences analysis. An alternative approach to studying the effect of personal crisis experience on stock allocation is to employ difference-in-differences estimations. We perform two types of difference-in-differences estimation using panel data spanning five years before and five years after the crisis: one at yearly frequency and the other collapsed at pre- and post- periods. We regress stock allocation on Large loss dummy, *Post* dummy for the after crisis period, and the interaction of the two, in addition to controls at the individual level.[21] The results are presented in the online Appendix Table A.8. All coefficients for the interaction term Large loss dummy $\times$ *Post* are negative and significant, indicating that investors tend to allocate fewer financial assets to stocks after having experienced a large equity loss. This negative relation holds for different thresholds of large loss, whether using panel data at yearly frequency or at collapsed at pre- and post-crisis periods, and regardless of whether 2008 is included in the sample. However, it is important to note that, unlike the main analysis, where we study different horizons from one to five years after the crisis, in the difference-in-differences analysis we focus on the averaged yearly change in stock allocation before and after the crisis. Nevertheless, the difference-in-differences results are consistent with our previous conclusion that the personal crisis experience affects investors' equity allocation decision.

6 Conclusion

In this paper, we explore how personal experiences affect individuals' future financial decisions. Using detailed administrative data covering the universe of Norwegian investors' financial securities' holdings and transactions, we measure individual-level investment experiences during the 2008 financial crisis. We then track these individuals over the subsequent five years and show that the idiosyncratic crisis experiences indeed shape investors' future investment behavior.

[21]For the panel data at yearly frequency, the dependent variable *StockAllocation* is the end-of-year value of stock holdings as a fraction of the end-of-year financial assets, multiplied by 100. For the collapsed difference-in-differences estimation, the dependent variable and controls are collapsed into a "pre" and "post"-period. For years before 2007, we have limited information on transactions and holdings, thus we can only obtain a coarse measure based on the end-of-year values of stock holdings. Unlike our measure in the main analysis, here we cannot distinguish between active changes in stock allocation and passive changes due to market value fluctuations. The details are presented in Online Appendix H

We find that individuals who suffered large equity losses during the 2008 financial crisis subsequently reduced their equity allocation and increased their portfolio concentration. These effects are not short-lived—they remain evident even five years after the crisis—and are robust to a variety of empirical specifications. Our results hold across different thresholds for defining large losses, in a matched sample of siblings that controls for unobservable personal traits, and under alternative measures of equity losses. The sustained influence of these effects indicates that idiosyncratic crisis experiences, beyond observable characteristics or broad economic trends, are fundamental to financial decision-making.

Consistent with previous literature, we also find that individuals' earlier exposure (ages 11 to 20) to stock market conditions influences investment decisions later in life. In addition, idiosyncratic crisis experience and formative year experience together shape future investment decisions. This reinforces the view that investment preferences are not fixed, but evolve through the accumulation of salient life events.

The findings of our paper support the belief that personal experiences significantly influence subsequent individual investment decisions and provide empirical evidence to quantify the magnitude and duration of these effects. These findings challenge the traditional rational investor model, underscoring the importance of unique personal experiences in economic decision-making. Our results have implications not only for theoretical modeling but also for financial advisory services and policy design. Tailoring advice and interventions to reflect investor lived experiences may improve both individual outcomes and market stability.

References

Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.

Andersen, S., Hanspal, T., and Nielsen, K. M. (2019). Once bitten, twice shy: The power of personal experiences in risk taking. *Journal of Financial Economics*, 132(3):97–117.

Barber, B. M. and Odean, T. (2001). Boys will be boys: Gender, overconfidence, and common stock investment. *The quarterly journal of economics*, 116(1):261–292.

Bateman, H., Islam, T., Louviere, J., Satchell, S., and Thorp, S. (2011). Retirement investor risk tolerance in tranquil and crisis periods: experimental survey evidence. *Journal of Behavioral Finance*, 12(4):201–218.

Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *The Quarterly journal of economics*, 119(1):249–275.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2022). Overreaction and diagnostic expectations in macroeconomics. *Journal of Economic Perspectives*, 36(3):223–244.

Brown, J. R., Ivković, Z., Smith, P. A., and Weisbenner, S. (2008). Neighbors matter: Causal community effects and stock market participation. *The Journal of Finance*, 63(3):1509–1531.

Bucher-Koenen, T. and Ziegelmeyer, M. (2014). Once burned, twice shy? financial literacy and wealth losses during the financial crisis. *Review of Finance*, 18(6):2215–2246.

Bugge, S. A., Guttormsen, H. J., Molnár, P., and Ringdal, M. (2016). Implied volatility index for the norwegian equity market. *International Review of Financial Analysis*, 47:133–141.

Calvet, L. E., Campbell, J. Y., and Sodini, P. (2007). Down or out: Assessing the welfare costs of household investment mistakes. *Journal of Political Economy*, 115(5):707–747.

Campbell, J. Y. (2006). Household finance. *The journal of finance*, 61(4):1553–1604.

Campbell, J. Y. and Viceira, L. M. (2002). *Strategic asset allocation: portfolio choice for long-term investors*. Clarendon Lectures in Economic.

Cesarini, D., Dawes, C. T., Johannesson, M., Lichtenstein, P., and Wallace, B. (2009). Genetic variation in preferences for giving and risk taking. *The Quarterly Journal of Economics*, 124(2):809–842.

Cesarini, D., Johannesson, M., Lichtenstein, P., Sandewall, Ö., and Wallace, B. (2010).

Genetic variation in financial decision-making. *The Journal of Finance*, 65(5):1725–1754.

Choi, J. J. and Robertson, A. Z. (2020). What matters to individual investors? evidence from the horse's mouth. *The Journal of Finance*, 75(4):1965–2020.

Cochrane, J. H. (2011). Presidential address: Discount rates. *The Journal of finance*, 66(4):1047–1108.

Fagereng, A., Gottlieb, C., and Guiso, L. (2017). Asset market participation and portfolio choice over the life-cycle. *The Journal of Finance*, 72(2):705–750.

Giannetti, M. and Wang, T. Y. (2016). Corporate scandals and household stock market participation. *The Journal of Finance*, 71(6):2591–2636.

Greenwood, R. and Shleifer, A. (2014). Expectations of returns and expected returns. *The Review of Financial Studies*, 27(3):714–746.

Guiso, L., Sapienza, P., and Zingales, L. (2018). Time varying risk aversion. *Journal of Financial Economics*, 128(3):403–421.

Guiso, L. and Sodini, P. (2013). Household finance: An emerging field. In *Handbook of the Economics of Finance*, volume 2, pages 1397–1532. Elsevier.

Haliassos, M. and Bertaut, C. C. (1995). Why do so few hold stocks? *the economic Journal*, 105(432):1110–1129.

Hansen, L. P. and Hodrick, R. J. (1980). Forward exchange rates as optimal predictors of future spot rates: An econometric analysis. *Journal of political economy*, 88(5):829–853.

Hoffmann, A. O., Post, T., and Pennings, J. M. (2013). Individual investor perceptions and behavior during the financial crisis. *Journal of Banking & Finance*, 37(1):60–74.

Hong, H., Kubik, J. D., and Stein, J. C. (2004). Social interaction and stock-market participation. *The journal of finance*, 59(1):137–163.

Hoopes, J. L., Langetieg, P., Nagel, S., Reck, D., Slemrod, J., and Stuart, B. A. (2022). Who sells during a crash? evidence from tax return data on daily sales of stock. *The Economic Journal*, 132(641):299–325.

Knüpfer, S., Rantapuska, E., and Sarvimäki, M. (2017). Formative experiences and portfolio choice: Evidence from the finnish great depression. *The Journal of Finance*, 72(1):133–166.

Malmendier, U. (2021). Experience effects in finance: Foundations, applications, and future directions. *Review of Finance*, 25(5):1339–1363.

Malmendier, U. and Nagel, S. (2011). Depression babies: do macroeconomic experiences affect risk taking? *The Quarterly Journal of Economics*, 126(1):373–416.

Malmendier, U. and Nagel, S. (2016). Learning from inflation experiences. *The Quarterly Journal of Economics*, 131(1):53–87.

Malmendier, U., Pouzo, D., and Vanasco, V. (2020a). Investor experiences and financial market dynamics. *Journal of Financial Economics*, 136(3):597–622.

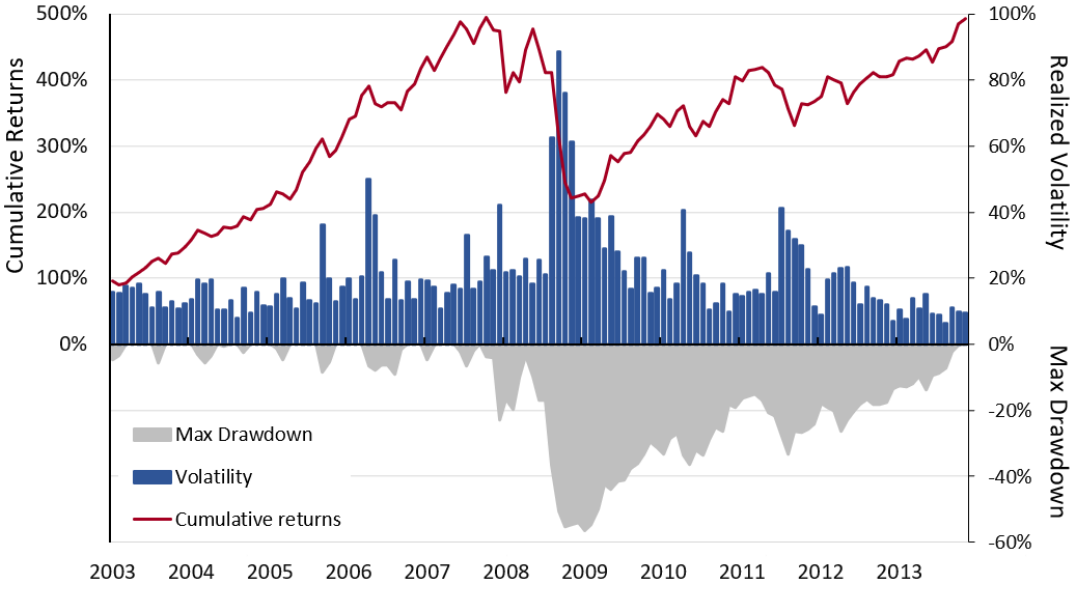
Malmendier, U., Pouzo, D., and Vanasco, V. (2020b). Investor experiences and international capital flows. *Journal of International Economics*, 124:103302.

Mankiw, N. G. and Zeldes, S. P. (1991). The consumption of stockholders and nonstockholders. *Journal of financial Economics*, 29(1):97–112.

Figure 1: Cumulative returns, maximum drawdown, and volatility

Panel A shows the cumulative returns, maximum drawdown, and monthly realized volatility for Oslo Stock Exchange All-Share Index (OSEAX) between January 2003 and December 2013. Cumulative returns are calculated using the monthly returns of OSEAX. The maximum drawdown is measured as the difference between the highest peak and the lowest trough values of OSEAX. The realized volatility is based on the monthly volatility of OSEAX. Panel B shows the implied volatility index, NOVIX, for the Norwegian equity market, both in logs and levels at a monthly frequency from January 2007 to December 2013.

Panel A: Cumulative returns, maximum drawdown, and realized volatility



Panel B: Implied Volatility (January 1, 2007 – December 31, 2013)

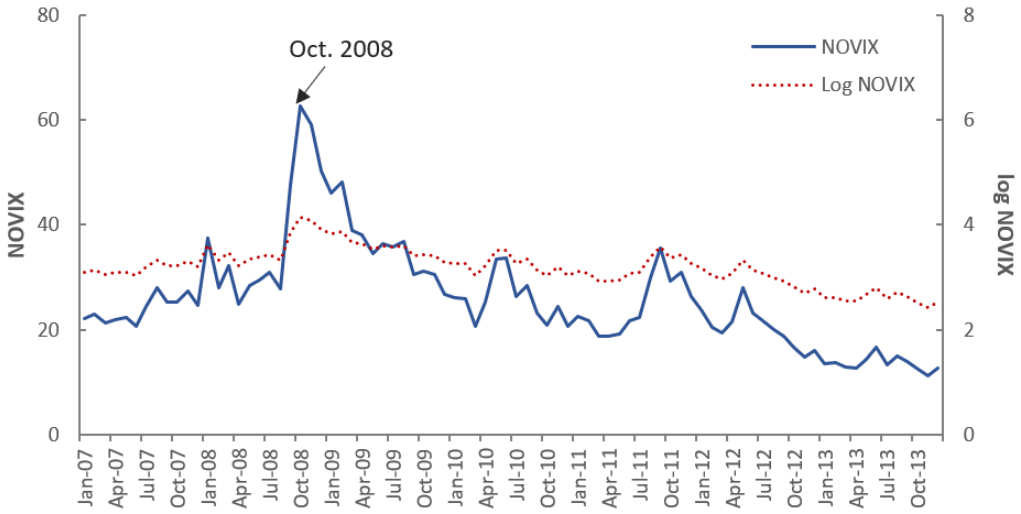


Table 1: **Summary Statistics (N = 344,209)**

This table reports descriptive statistics related to stock allocation, stock concentration, and allocation in other financial assets for the 1-year, 3-year, and 5-year windows (Panel A), experience-related variables and sociodemographic characteristics (Panel B). The definitions of all variables are described in Section 2.3. Please also refer to Equation (1) for *Change in stock allocation (%)*, Equation (2) for *Change in stock concentration (ΔHHI)*, Equation (3) for *Change in cash deposits (%)*, *Change in mutual funds (%)*, *Change in bonds and money market funds (%)*, and *Change in unlisted securities (%)*. Equation (4) defines *Large loss dummy*, and Equation (6) defines *Formative-year experience, FYE ($\lambda = 0 or 1.5$)*.

Panel A: Allocation in stocks and other financial assets

	1-year		3-year		5-year	
	Mean	Std	Mean	Std	Mean	Std
Stock allocation and concentration						
Change in stock allocation (%)	-2.12	13.36	-0.07	21.55	0.17	28.02
Liquidation (dummy)	0.14	0.35	0.21	0.41	0.27	0.45
Change in the number of stocks (Δlog)	-0.03	0.31	-0.08	0.44	-0.14	0.51
Change in stock concentration (ΔHHI)	-0.58	13.83	0.03	18.71	1.07	20.74
Allocation in other financial assets (%)						
Change in cash deposits	2.0	42.2	9.6	53.2	16.8	58.2
Change in mutual funds	0.0	6.0	-0.1	8.2	0.0	10.3
Change in bonds & money market funds	4.5	14.0	3.4	17.4	7.4	24.2
Change in unlisted securities	0.7	18.9	1.6	23.2	2.6	25.9

Panel B: Experience, wealth, and sociodemographic variables

	Mean	Median	Std.
Experience related variables			
Large loss dummy (threshold: 90th percentile)	0.10	0.00	0.30
Large loss dummy (threshold: 75th percentile)	0.25	0.00	0.44
Formative-year experience, FYE ($\lambda = 0$)	4.6%	3.2%	5.1%
Formative-year experience, FYE ($\lambda = 1.5$)	4.8%	3.8%	7.0%
Wealth and sociodemographic characteristics			
Personal wealth in 2007 ('000 NOK)	1,939	784	16,012
Annual income in 2007('000 NOK)	415	380	560
Personal liabilities in 2007 ('000 NOK)	904	350	3,614
Dividend income in 2007 ('000 NOK)	22	0.402	378
Social security recipients in 2007 (dummy)	0.25	0	0.43
Ownership in private firms in 2007 (dummy)	0.11	0	0.31
Self-employed in 2007 (dummy)	0.15	0	0.36
Age in 2007	52	52	15.9
Male in 2007 (dummy)	0.71	1	0.45
Married in 2007 (dummy)	0.71	1	0.46
College degree in 2007 (dummy)	0.43	0	0.50
Econ/Business education in 2007 (dummy)	0.10	0	0.30

Table 2: **Stock portfolio of individual investors in 2008**

This table presents summary statistics for the characteristics of investors' stock portfolios in 2008. For each investor, we construct *Number of stocks held*, which is the number of unique stocks held at the end of year 2008. *Value of stocks held* is the total value of all stocks at the end of 2008. *Number of trades* is the number of transactions conducted by an investor during 2008. *Value of transactions* is the sum of all transactions (the absolute value of stock purchases and sales). *Total loss* and *Realized loss* are defined in Equation (4) and Subsection 2.3.3. We further scale these two variables by the investor's total wealth at the end of 2007 to get *Total loss%* and *Realized loss%*, where personal wealth includes stocks portfolio as well as bank deposits, mutual funds, bonds and money market funds, other unlisted securities, and real capital.

	Obs.	Mean	Std.	p10	p25	p50	p75	p90
Stock portfolio characteristics (values are shown in thousand NOK)								
Number of stocks held	344,209	2.1	2.6	1	1	1	3	5
Value of stocks held	344,209	1539	18200	52	137	376	1001	2406
Number of transactions	175,107	3.7	5.4	1	1	2	4	9
Value of transactions	175,107	4431	33600	4	16	56	277	1474
Loss in thousand NOK								
Total loss	344,209	160	1,727	0.5	5	23	94	306
Realized loss	175,107	224	2,349	-2.2	5	40	152	460
Loss scaled by personal wealth in 2007								
Total loss%	344,209	12 %	28 %	0.2 %	1.0 %	4 %	13 %	33 %
Realized loss%	175,107	16 %	37 %	-0.3 %	1.1 %	7 %	20 %	46 %

Table 3: **Change in stock allocation after the crisis experience**

This table reports changes in individuals' stock allocations after the crisis experience in 2008. The dependent variable is *N-year change in stock allocation* (where N is 1, 3, or 5) as defined in Equation (1). *Large loss dummy* equals one if *Total loss%* falls above the 90th or the 75th percentiles, respectively. Standard errors reported in parentheses are clustered by the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Dependent Variable:				Large Losses Dummy using Different Thresholds:					
N-year change in				Above the 90th Percentile			Above the 75th Percentile		
stock allocation	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Large loss dummy				-3.40***	-4.64***	-4.43***	-2.75***	-2.48***	-2.60***
				(0.27)	(0.56)	(0.55)	(0.11)	(0.34)	(0.41)
Personal wealth	0.52***	-0.05	0.23**	0.39***	-0.23**	0.05	0.36***	-0.20**	0.07
	(0.04)	(0.10)	(0.11)	(0.04)	(0.09)	(0.11)	(0.04)	(0.09)	(0.11)
Annual salary	-0.07***	0.45***	0.58***	-0.08***	0.44***	0.57***	-0.07***	0.45***	0.58***
	(0.02)	(0.06)	(0.08)	(0.02)	(0.06)	(0.08)	(0.02)	(0.06)	(0.08)
Personal liabilities	-0.05***	0.13***	0.18***	-0.05***	0.12***	0.17***	-0.05***	0.12***	0.17***
	(0.01)	(0.02)	(0.03)	(0.01)	(0.02)	(0.03)	(0.01)	(0.02)	(0.03)
Dividend income	-0.47***	-0.43***	-0.70***	-0.33***	-0.24***	-0.52***	-0.22***	-0.21***	-0.47***
	(0.04)	(0.07)	(0.08)	(0.03)	(0.07)	(0.08)	(0.03)	(0.07)	(0.08)
Social security recipient	0.26***	0.88***	1.22***	0.20***	0.81***	1.15***	0.18***	0.81***	1.15***
	(0.06)	(0.17)	(0.20)	(0.06)	(0.17)	(0.21)	(0.06)	(0.17)	(0.21)
Ownership in private firm	1.08***	-0.18	-0.42	0.96***	-0.34*	-0.58**	0.79***	-0.44**	-0.69***
	(0.06)	(0.20)	(0.27)	(0.06)	(0.19)	(0.26)	(0.06)	(0.18)	(0.25)
Self-employed	0.36***	-1.25***	-1.72***	0.34***	-1.28***	-1.74***	0.27***	-1.33***	-1.80***
	(0.06)	(0.19)	(0.25)	(0.06)	(0.18)	(0.25)	(0.06)	(0.18)	(0.24)
Age	-0.63***	-2.17***	-3.02***	-0.94***	-2.59***	-3.42***	-1.03***	-2.53***	-3.39***
	(0.13)	(0.35)	(0.53)	(0.15)	(0.35)	(0.52)	(0.16)	(0.36)	(0.54)
Male	-0.20**	-0.09	-0.36***	-0.06	0.10	-0.17	-0.06	0.04	-0.22
	(0.08)	(0.09)	(0.13)	(0.09)	(0.09)	(0.13)	(0.08)	(0.09)	(0.13)
Married	-0.09	0.19	0.29**	-0.14	0.12	0.23	-0.13**	0.16	0.26**
	(0.06)	(0.11)	(0.13)	(0.06)	(0.11)	(0.13)	(0.06)	(0.11)	(0.13)
College degree	-0.37***	0.77***	1.25***	-0.35***	0.80***	1.28***	-0.32***	0.82***	1.30***
	(0.06)	(0.16)	(0.24)	(0.06)	(0.15)	(0.23)	(0.06)	(0.15)	(0.23)
Econ/Business education	-0.10	-0.64***	-0.39**	-0.09	-0.63***	-0.38**	-0.10	-0.64***	-0.39**
	(0.08)	(0.16)	(0.19)	(0.08)	(0.16)	(0.19)	(0.08)	(0.16)	(0.19)
Intercept	-1.93***	6.01***	6.94***	0.37***	9.13***	9.92***	1.11	8.73***	9.79***
	(0.59)	(1.51)	(2.17)	(0.70)	(1.54)	(2.15)	(0.71)	(1.65)	(2.33)
Sample size	344,209	336,623	329,016	344,209	336,623	329,016	344,209	336,623	329,016
Adjusted R^2	0.004	0.007	0.007	0.009	0.011	0.009	0.011	0.009	0.009

Table 4: **Stock concentration - extensive and intensive margins**

Panel A reports the results from Logit model on the likelihood of investors exiting the stock market entirely (the extensive margin). Panel B examines how investors adjust their portfolio composition conditional on remaining in the market (the intensive margin). The sample in Panel B excludes all investors who completely liquidated their stock holdings. The coefficients of control variables are not tabulated in this table, but all the control variables are included in the regressions. Standard errors reported in parentheses are clustered by the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A: Stock liquidation with Logit model (extensive margin)

Dependent Variable	**Large Loss Dummy using Different Thresholds:**					
Stock liquidation	Above the 90th Percentile			Above the 75th Percentile		
N years after the crisis	1 Year	3 Year	5 Year	1 Year	3 Year	5 Year
Large loss dummy	0.68***	0.60***	0.57***	0.52***	0.45***	0.45***
	(0.08)	(0.05)	(0.03)	(0.06)	(0.05)	(0.03)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Pseudo R^2	0.43	0.23	0.16	0.43	0.23	0.16
Probability of stock liquidation given all controls estimated at mean and ...						
Large loss dummy = 0	0.037	0.124	0.195	0.035	0.119	0.187
Large loss dummy = 1	0.070	0.205	0.301	0.057	0.175	0.265
Marginal effects	0.033	0.081	0.106	0.022	0.056	0.078

Panel B: Stock concentration among continuing investors (intensive margin)

	Large Loss Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in the number of stocks						
Large loss dummy	-0.04***	-0.08***	-0.11***	-0.03***	-0.06***	-0.09***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R-squared	0.003	0.007	0.009	0.004	0.008	0.012
Dependent Variable: Change in stock concentration index after the crisis experience						
Large loss dummy	1.59***	4.71***	5.85***	1.23***	4.50***	5.38***
	(0.21)	(0.25)	(0.32)	(0.14)	(0.18)	(0.24)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R^2	0.003	0.007	0.009	0.003	0.012	0.014

Table 5: **Analysis using the siblings subsample**

This table presents analysis using a subsample of matched sibling pairs. Panel A reports descriptive statistics for the siblings that are divided into groups—those who experienced a large loss during the crisis and those who did not. The differences are shown in the last two columns based on a two-sample t-test (for means) and Mann-Whitney U rank-sum test (for medians). In Panel B, siblings fixed effects are added to the baseline models, as specified in estimation equations (7), (8), (9), and (10). Standard errors clustered at the sibling-pair level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A: Summary statistics on the sibling sample

Variables	Large loss dummy (above the 90th Percentile)					
	dummy = 1		dummy = 0		Difference (p-val)	
	Mean	Median	Mean	Median	Mean	Median
Personal wealth in 2007 ('000 NOK)	1703	813	2776	832	0.02	0.00
Annual salary in 2007 ('000 NOK)	626	507	540	476	0.00	0.00
Personal liabilities in 2007 ('000 NOK)	1175	541	1301	634	0.26	0.76
Social security recipient (dummy)	0.09	0	0.10	0	0.06	0.06
Ownership in private firm (dummy)	0.13	0	0.16	0	0.00	0.00
Self-employed (dummy)	0.12	0	0.18	0	0.00	0.00
Age	45.6	46	46.2	47	0.01	0.02
Male	0.77	1	0.68	1	0.00	0.00
Married	0.70	1	0.72	1	0.05	0.05
College degree	0.53	1	0.53	1	0.85	0.85
Econ/Business education	0.13	0	0.11	0	0.00	0.00
Number of individuals	5,266		5,535			

Variables	Large loss dummy (above the 75th Percentile)					
	dummy = 1		dummy = 0		Difference (p-val)	
	Mean	Median	Mean	Median	Mean	Median
Personal wealth in 2007 ('000 NOK)	1980	832	2629	876	0.03	0.00
Annual salary in 2007 ('000 NOK)	611	516	551	477	0.00	0.00
Personal liabilities in 2007 ('000 NOK)	1167	519	1134	611	0.55	0.03
Social security recipient (dummy)	0.10	0	0.11	0	0.00	0.00
Ownership in private firm (dummy)	0.13	0	0.17	0	0.00	0.00
Self-employed (dummy)	0.13	0	0.19	0	0.00	0.00
Age	47.1	48	47.4	48	0.03	0.06
Male	0.73	1	0.68	1	0.00	0.00
Married	0.73	1	0.74	1	0.02	0.02
College degree	0.54	1	0.51	1	0.00	0.00
Econ/Business education	0.13	0	0.12	0	0.00	0.00
Number of individuals	11,565		10,746			

Panel B: Stock allocation and concentration after the crisis

	Large losses dummy using different thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation						
Large loss dummy	-2.51***	-3.63***	-4.46***	-1.52***	-1.46**	-1.71*
	(0.76)	(1.23)	(1.60)	(0.43)	(0.74)	(0.96)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Pre-crisis stock allocation FE	Yes	Yes	Yes	Yes	Yes	Yes
Sibling fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	10,801	10,801	10,801	22,311	22,311	22,311
Adjusted R^2	0.190	0.152	0.152	0.168	0.141	0.155
Dependent Variable: Stock liquidation N-year after the crisis (Logit)						
Large loss dummy	0.42**	0.40***	0.39***	0.22*	0.15*	0.16**
	(0.18)	(0.12)	(0.10)	(0.13)	(0.08)	(0.07)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Pre-crisis stock allocation FE	Yes	Yes	Yes	Yes	Yes	Yes
Sibling fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	1,729	2,557	3,144	3,601	5,275	6,449
Pseudo R^2	0.567	0.283	0.209	0.580	0.305	0.209
Dependent Variable: N-year change in the number of stocks (conditional on staying in the market)						
Large loss dummy	-0.04***	-0.08***	-0.11***	-0.03***	-0.06***	-0.09***
	(0.012)	(0.018)	(0.022)	(0.008)	(0.012)	(0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Pre-crisis stock allocation FE	Yes	Yes	Yes	Yes	Yes	Yes
Sibling fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	9,700	9,123	8,586	20,016	18,851	17,826
Adjusted R^2	0.131	0.125	0.135	0.126	0.150	0.155
Dependent Variable: N-year change in concentration index (conditional on staying in the market)						
Large loss dummy	1.86***	4.52***	6.25***	1.98***	5.22***	6.18***
	(0.73)	(1.06)	(1.23)	(0.50)	(0.72)	(0.86)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Pre-crisis stock allocation FE	Yes	Yes	Yes	Yes	Yes	Yes
Sibling fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	9,700	9,123	8,586	20,016	18,851	17,826
Adjusted R^2	0.126	0.114	0.114	0.131	0.136	0.122

Table 6: **Changes in the allocation of other financial assets**

This table replicates Table 3, replacing dependent variables with 1-, 3-, and 5-year changes in cash deposits, mutual funds, bonds and money market funds, and other unlisted securities. The coefficients of control variables are not tabulated in this table, but all the control variables specified in Equation (7) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Losses Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent variable: Changes in cash deposits						
Large loss dummy	2.19***	1.08*	0.32	1.83***	1.26***	1.19**
	(0.54)	(0.63)	(0.80)	(0.38)	(0.40)	(0.57)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	343,511	335,951	328,366	343,511	335,951	328,366
Adjusted R^2	0.012	0.024	0.030	0.012	0.024	0.030
Dependent variable: Changes in mutual funds						
Large loss dummy	-0.81***	-0.97***	-2.50***	-0.23**	-0.52***	-1.24***
	(0.18)	(0.19)	(0.29)	(0.10)	(0.10)	(0.15)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	343,511	335,951	328,366	343,511	335,951	328,366
Adjusted R^2	0.011	0.007	0.012	0.011	0.007	0.012
Dependent variable: Changes in bonds and money market funds						
Large loss dummy	0.01	0.09**	0.26***	0.04	0.13***	0.35***
	(0.04)	(0.04)	(0.05)	(0.02)	(0.04)	(0.05)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	343,511	335,951	328,366	343,511	335,951	328,366
Adjusted R^2	0.0002	0.001	0.002	0.0002	0.001	0.002
Dependent variable: Changes in unlisted securities						
Large loss dummy	-0.14	0.24*	0.14	-0.08	0.09	-0.04
	(0.12)	(0.14)	(0.15)	(0.07)	(0.10)	(0.10)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	343,511	335,951	328,366	343,511	335,951	328,366
Adjusted R^2	0.001	0.005	0.009	0.001	0.005	0.009

Table 7: **Potential financial impact of the crisis experience**

This table reports the potential financial impact of the crisis experience. For each individual, we calculate *Difference* as the difference between two values: i) the end-of-year portfolio value plus income from all stock sales minus the expense from all stock purchases during the period; and ii) the buy-and-hold value of the portfolio held at the end of 2008. The differences are shown in the last two columns based on the two-sample t-test (for means) and Mann-Whitney U rank-sum test (for medians).

	Large loss = 1		Large loss = 0		Difference (p-val)	
Difference	Mean	Median	Mean	Median	Mean	Median
Large loss dummy (above the 90th percentile)						
1-Year	-188,520	-71,862	-23,140	-2,822	0.00	0.00
3-Year	-265,583	-95,064	-39,191	-3,889	0.00	0.00
5-Year	-149,496	-50,007	-11,984	-202	0.00	0.00
N	27,701	27,701	267,060	267,060		
Large loss dummy (above the 75th percentile)						
1-Year	-219,721	-111,830	-23,167	-3,327	0.00	0.00
3-Year	-293,780	-122,678	-38,918	-3,544	0.00	0.00
5-Year	-174,986	-67,676	-12,815	-211	0.00	0.00
N	71,138	71,138	222,550	222,550		

Table 8: **Imputed expected returns and expectations of returns**

This table reports the dividend yield-implied expected stock market returns and the expectations of future stock returns at the end of 2008. In panel A, the the dividend yield-implied expected stock market returns over the one- to five-year windows are estimated using the dividend yield model from Cochrane (2011). The regression equation is $R^e_{t:t+k} = \alpha + \beta \times D_t/P_t + \epsilon_{t+k}$. The dependent variable $R^e_{t:t+k}$ is the stock market return (from Datastream OSE Industry total return index) less the 1-month government bond rate (that is based on government securities and NIBOR). The variable D_t/P_t is dividend yield. Data are annual for the period of 1970–2008. The regression t-statistic uses the Hansen–Hodrick (1980) correction for two to five year horizons. $\sigma[E_t(R_e)]$ represents the standard deviation of the fitted value, $\sigma(\hat{\beta}) \times D_t/P_t$. Risk-free rate for 1-, 3-, 5-year horizons are the government bond yield for the maturity of 1 year, 3 years, and 5 years, respectively. The risk-free rate for 2- and 4-year horizons are extrapolated based on the 1-, 3-, and 5- year rates. In panel B, the expectations of future stock returns are imputed using the models from Greenwood and Shleifer (2014) where they regress survey expectations of future stock market returns on past stock market returns and price-dividend ratio. We replace the stock market returns with the personal past 12-month returns at the end of 2008 and estimate the expected one-year returns for each individual investor. Model 1 is $Exp_t = 5.789 + 3.133R_{t-12:t-1}$ and model 2 is $Exp_t = -7.979 + 3.354R_{t-12:t-1} + 3.404log(P_t/D_t) + \mu$. The estimated returns are truncated at -100% and 100%.

Panel A: Dividend yield-implied expected stock market returns

Horizon (Year)	1	2	3	4	5
β	28.7	41.0	41.7	33.9	32.4
$t(\beta)$	2.0	2.2	2.7	3.5	3.7
R^2	0.23	0.30	0.30	0.19	0.13
$\sigma[E_t(R_e)]$	0.32	0.46	0.46	0.38	0.36
$\sigma[E_t(R_e)]/E_t(R_e)$	2.48	1.62	1.14	0.72	0.51
Risk-free rate	4.9%	4.7%	4.6%	4.6%	4.7%
The forecast future return for the next 5 years:					
Imputed excess return	74.2%	115.7%	129.8%	124.6%	139.9%
Imputed gross return	79.0%	120.4%	134.4%	129.2%	144.6%

Panel B: Expectations of future stock returns

		Large loss dummy using different thresholds:			
		Above the 90th Percentile		Above the 75th Percentile	
		Dummy=1	Dummy=0	Dummy=1	Dummy=0
Model 1	N	32,831	316,161	82,077	266,915
	Mean	-99.9%	-4.9%	-76.5%	5.4%
	Median	-100.0%	-2.7%	-81.5%	15.4%
Model 2	N	32,831	316,161	82,077	266,915
	Mean	-100.0%	-11.4%	-82.4%	-42.0%
	Median	-100.0%	-9.0%	-93.4%	-6.0%

Table 9: **Formative-year experience and the crisis experience**

This table reports the results on formative-year experience as specified in Equation (12). Formative-year returns (*FYE*) are defined in Equation (6) with $\lambda = 1.5$. The coefficients of control variables are not tabulated, but all control variables are included in the regressions. Standard errors reported in parentheses are clustered by the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Loss Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation						
Large loss dummy	-3.98***	-5.14***	-4.74***	-3.04***	-2.95***	-3.13***
	(0.20)	(0.61)	(0.57)	(0.10)	(0.37)	(0.47)
FYE	-0.66	3.67***	5.07***	-0.89**	2.46***	3.27***
	(0.47)	(1.03)	(1.48)	(0.44)	(0.77)	(1.23)
Large loss dummy × *FYE*	10.09***	8.72**	5.52**	5.54***	8.87**	9.93**
	(3.25)	(4.11)	(5.52)	(1.46)	(3.56)	(4.62)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Adjusted R^2	0.009	0.011	0.009	0.011	0.009	0.009
Dependent Variable: Stock liquidation N years after the crisis (Logit)						
Large loss dummy	0.62***	0.56***	0.52***	0.45***	0.41***	0.40***
	(0.04)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)
FYE	0.37**	0.19	0.16	0.38**	0.24*	0.21*
	(0.15)	(0.13)	(0.11)	(0.16)	(0.13)	(0.12)
Large loss dummy × *FYE*	0.04	0.06	0.25	-0.10	-0.25	-0.18
	(0.37)	(0.30)	(0.25)	(0.23)	(0.21)	(0.20)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Pseudo R^2	0.400	0.195	0.134	0.400	0.195	0.134
Dependent Variable: N-year change in the number of stocks (conditional on staying in the market)						
Large loss dummy	-0.03***	-0.08***	-0.10***	-0.02***	-0.06***	-0.08***
	(0.003)	(0.005)	(0.01)	(0.002)	(0.003)	(0.004)
FYE	-0.02***	-0.04***	-0.03**	-0.01	-0.02	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Large loss dummy × *FYE*	-0.05*	-0.06	-0.12*	-0.07***	-0.10***	-0.12***
	(0.03)	(0.05)	(0.06)	(0.02)	(0.03)	(0.04)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R^2	0.003	0.007	0.010	0.004	0.008	0.012
Dependent Variable: N-year change in concentration index (conditional on staying in the market)						
Large loss dummy	1.43***	4.67***	5.59***	1.00***	4.26***	5.03***
	(0.21)	(0.28)	(0.35)	(0.12)	(0.18)	(0.24)
FYE	1.68***	3.26***	0.89	0.82	1.97***	0.89
	(0.45)	(0.79)	(0.89)	(0.45)	(0.80)	(0.89)
Large loss dummy × *FYE*	2.99*	0.73	2.31***	4.44***	4.80***	6.97***
	(1.66)	(2.44)	(0.86)	(0.02)	(1.70)	(2.09)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R^2	0.003	0.008	0.009	0.003	0.012	0.014

Internet Appendix

for

Finance Gets Personal: The Lasting Effects of Idiosyncratic Crisis Experiences on Investment Choices

Appendix A: Pre-crisis characteristics and the likelihood of large loss

Investor preferences shape pre-crisis asset allocation decisions, which may in turn influence outcomes during financial downturns. To examine how investors' pre-crisis characteristics and portfolio composition relate to the likelihood of experiencing substantial losses, we estimate the following Logit model:

$$\text{Large loss dummy}_{i,2008} = \alpha + \beta \text{Portfolio Characteristics}_{i,2007} + \gamma Investor\ Characteristics_{i,2007} + \epsilon_i \tag{14}$$

where i indexes the individual investor. The dependent variable, *Large loss dummy*, is a binary indicator equal to one if an investor experienced a large equity loss in 2008. The specific definition of this variable, based on two alternative thresholds, is provided in Subsection 2.3.3.

We measure portfolio characteristics using the following three variables. *Stock portfolio value* is the natural logarithm of total value of stock portfolio at the end of 2007. *Number of stocks* is the natural logarithm of the number of unique stocks held at the end of 2007. *Stock concentration* is the Herfindahl-Hirschman Index *HHI* which measures the sum of the squared weights of individual stocks in the portfolio by the end of 2007.

Investor characteristics include the following variables. Wealth group dummies based on quartiles of the wealth distribution (with the first quartile as the reference group). Age group dummies for investors under 35 years of age (reference group), 35-49, 50-64, and 65 years or older. *Married* is a dummy variable that equals one if a person is married or entered into a registered partnership, and zero otherwise. *College degree* equals one if a person's highest degree level is in college / university or higher, and otherwise zero. *Econ/Business education* is a dummy variable that takes a value of one if an investor received a college / university degree in the field of business administration, economics, or finance, and otherwise zero.

The estimation results are presented in Table A.1. They suggest that investors with larger stock holdings, lower wealth, younger age (under 35), male gender, and single marital status were more likely to incur large losses in 2008.

Table A.1: Likelihood of experiencing a large loss

This table replicates Table 3 and 4 using the *Total loss%* as an alternative measure for the large losses dummy. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Dependent Variable:	Large Losses Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	(1)	(2)	(3)	(4)	(5)	(6)
Stock portfolio value	0.63		1.15	0.72		1.27
	(0.09)		(0.08)	(0.08)		(0.07)
Number of stocks	0.31		0.58	0.29		0.62
	(0.06)		(0.06)	(0.06)		(0.04)
Stock concentration	-0.72		-0.90	-0.44		-0.72
	(0.11)		(0.08)	(0.11)		(0.07)
Wealth group [p25, p50)		-0.88	-1.71		-0.61	-1.64
		(0.12)	(0.16)		(0.09)	(0.16)
Wealth group [p50, p75)		-0.82	-2.38		-0.58	-2.43
		(0.11)	(0.13)		(0.08)	(0.14)
Wealth group [p75, 100]		-0.56	-3.50		-0.45	-3.65
		(0.11)	(0.19)		(0.09)	(0.17)
Age group 35-49		-0.24	-0.31		-0.19	-0.29
		(0.19)	(0.29)		(0.13)	(0.26)
Age group 50-64		-0.53	-0.68		-0.41	-0.64
		(0.19)	(0.29)		(0.13)	(0.27)
Age group 65+		-0.86	-1.17		-0.64	-1.07
		(0.20)	(0.30)		(0.15)	(0.28)
Male		0.38	0.49		0.20	0.35
		(0.03)	(0.04)		(0.03)	(0.04)
Married		-0.17	-0.08		-0.09	-0.02
		(0.03)	(0.04)		(0.02)	(0.03)
College degree		0.06	-0.13		0.14	-0.06
		(0.05)	(0.04)		(0.03)	(0.03)
Business education		0.00	0.01		-0.03	0.02
		(0.02)	(0.04)		(0.02)	(0.03)
Intercept	-9.67	-1.56	-13.74	-9.36	-0.62	-13.60
	(1.12)	(0.23)	(1.14)	(1.02)	(0.18)	(0.94)
Observations	293,686	293,686	293,686	293,686	293,686	293,686
Pseudo R-squared	0.158	0.044	0.299	0.209	0.026	0.357

Appendix B: *Total losses%* as an alternative measure

To capture crisis experience, we primarily rely on the dummy variable *Large loss dummy*, based on the variable *Total loss%*, applying different thresholds to identify a large loss. The rationale is to focus on larger losses, as most investors experienced losses during the crisis year. We describe in detail on how we construct *Large loss dummy* in the data section. However, instead of dummy variables, we can also use continuous variable *Total loss%* directly in our regressions. Thus, we re-run all our baseline regressions Equation (7)-(10) by replacing *Large loss dummy* with *Total loss%*. The results are presented in Table A.2.

We include the same set of control variables as in the baseline regressions and report only the coefficient and standard errors of the main variable, *Total loss%*. We find similar results in terms of signs and significance of all coefficients. As shown in the top panel, a one-percentage-point increase in *Total loss%* is associated with a decline in the stock allocation by 5.72% over one year, by 6.75% over three years, and by 6.12% over five years. Given that the average stock allocation 22.3% at the end of 2007 and 17.3% at the end of 2008, a reduction of about 6% is substantial. For the likelihood of liquidation, all the coefficients on *Total loss%* are positive and significant, indicating that the greater the losses investors experienced in the financial crisis, the more likely they will liquidate their holdings of stocks in the future. Regarding the change in the number of stocks and the stock concentration index, a one-percentage-point increase in *Total loss%* is associated with a decrease in the number of stocks by 0.07 over one year to 0.21 over five years and an increase in stock concentration index by 3.23 over one year to 12.3 over five years.

All these findings indicate that the total loss in stock portfolio during the financial crisis, as a percentage of investors' pre-crisis wealth, has impact on future stock allocation and stock concentration for at least up to five years.

Table A.2: Analysis using Total loss%

This table replicates Table 3 and 4 using the *Total loss%* as an alternative measure for the large losses dummy. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation			
Total loss%	-5.72***	-6.75***	-6.12***
	(0.72)	(0.97)	(0.93)
Controls	Yes	Yes	Yes
Sample size	344,209	336,623	329,016
Adjusted R^2	0.011	0.011	0.009
Dependent Variable: Stock liquidation N-year after the crisis (Logit)			
Total loss%	0.81***	0.83***	0.80***
	(0.09)	(0.08)	(0.07)
Controls	Yes	Yes	Yes
Sample size	344,209	336,623	329,016
Pseudo R^2	0.43	0.22	0.16
Dependent Variable: N-year change in the number of stocks			
Total loss%	-0.07***	-0.15***	-0.21***
	(0.005)	(0.010)	(0.013)
Controls	Yes	Yes	Yes
Sample size	292,468	265,278	242,020
Adjusted R^2	0.005	0.010	0.015
Dependent Variable: N-year change in stock concentration			
Total loss%	3.23***	10.00***	12.30***
	(0.27)	(0.40)	(0.55)
Controls	Yes	Yes	Yes
Sample size	292,468	265,278	242,020
Adjusted R^2	0.004	0.013	0.015

Appendix C: Realized losses as an alternative measure

In the main analysis, our measure for crisis experience, *Large loss dummy*, is based on the total loss during the 2008 crisis which is the sum of the paper loss and the realized losses. The idea is that investors could already feel the pain with a loss on paper. However, one could argue that the paper loss and the realized loss are different and that investors feel more direct pain from the realized losses. Therefore, we carry out an additional analysis using the realized loss to construct the *Large loss dummy*.

We describe how we construct *Realized loss* in the data section. Please note that this measure is only available for a sub-sample of investors who made at least one transaction in 2008, meaning that investors who only experienced paper losses are not included in this analysis. The summary statistics of the realized loss are shown in Table 2. On average, *Realized loss* counts for 15.9% of financial asset. Similar to the analysis before, we apply two different thresholds to define large loss. Under the new definition, *Large loss dummy* equals one if *Realized loss%* lies above the 90th percentile (above 45.8%) and the 75th percentile (above 19.7%), respectively.

We re-run the same analysis as in Table 3 and Table 4 using this alternative measure. The regression results are shown in Table A.3. We only report the coefficients and standard errors of the main variable, *Large loss dummy*. Almost all coefficients are significant and exhibit the same signs as those based on *Total loss%* for the full sample; the magnitudes of the coefficients are also similar. All these findings indicate that investors who have experienced a large realized loss in the financial crisis are associated with a decrease in the stock allocation and the number of stocks in the portfolio and an increase in stock concentration in the years after. The impacts of the crisis experience are robust to the alternative measure using realized losses and the sub-sample of investors who traded during the crisis.

Table A.3: Analysis using realized losses

This table replicates Table 3 and 4 using *Realized loss%* as an alternative measure for the large losses dummy. This sample includes investors who traded stocks on Oslo Stock Exchange in 2008. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Losses Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation						
Large loss dummy	-3.31***	-6.49***	-5.89***	-2.63***	-3.57***	-3.13***
	(0.38)	(0.90)	(0.71)	(0.15)	(0.49)	(0.47)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	173,273	170,833	168,425	173,273	170,833	168,425
Adjusted R^2	0.005	0.013	0.011	0.006	0.012	0.010
Dependent Variable: Stock liquidation N-year after the crisis (Logit)						
Large loss dummy	0.17***	0.27***	0.28***	0.06	0.14***	0.12***
	(0.05)	(0.03)	(0.03)	(0.04)	(0.03)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	173,273	170,833	168,425	173,273	170,833	168,425
Pseudo R^2	0.450	0.275	0.206	0.450	0.275	0.206
Dependent Variable: N-year change in the number of stocks						
Large loss dummy	-0.04***	-0.08***	-0.09***	-0.03 ***	-0.07***	-0.08***
	(0.004)	(0.01)	(0.01)	(0.003)	(0.004)	(0.01)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	127,785	116,991	106,293	127,785	116,991	106,293
Adjusted R^2	0.003	0.005	0.007	0.004	0.007	0.009
Dependent Variable: N-year change in stock concentration						
Large loss dummy	1.67***	3.40***	3.56***	1.51***	3.57***	3.49***
	(0.27)	(0.40)	(0.55)	(0.20)	(0.22)	(0.32)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	127,785	116,991	106,293	127,785	116,991	106,293
Adjusted R^2	0.004	0.005	0.004	0.005	0.007	0.006

Appendix D: A different threshold to define large losses

In our main text, all the regression results are shown for the the *Large loss dummy* using the 90th and 75th percentiles of *Total loss%* to identify a substantial loss. In this section, we apply the median threshold as a robustness check, creating *Large loss dummy* equals one if *Total loss%* lies above the 50th percentile ($Total\,loss\% > 4\%$) and zero otherwise. As mentioned in the main text, the higher the threshold, the smaller the treated group and the larger the shock experienced by an investor. Thus, lowering the threshold to the median means that more investors are in the treated group, with on average fewer extreme losses within the treated group.

We re-run all our baseline regressions (Equations 7-10), applying the median *Total loss%* as the threshold for the *Large loss dummy*. The results are presented in Table A.4. The top panel shows that all the coefficients on *Large loss dummy* are negative and significant, after controlling for a rich set of individual characteristics such as age, personal wealth, employment status, marital status and education. An experience of 4% loss (the 90th percentile) in the equity portfolio during 2008 is associated with a decline in the equity allocation by 2.3%, 0.9%, 1.2% in one, three, five years after the 2008 crisis. Combined with the results in the main text, we observe that the magnitude of the coefficients on *Large loss dummy* is much larger using a stricter threshold (Table 3) than those using a moderate threshold (in Table A.4). This perhaps suggests that the larger the loss an investor experienced, the smaller the proportion of wealth she would allocate to stocks in the future, consistent with the results in Appendix A.

Regarding the likelihood of stock liquidation, all the coefficients on *Large loss dummy* are positive and significant, consistent with our main text. The magnitude of the coefficients are slightly smaller than those using the stricter thresholds. Again, when examining changes in the number of stocks and in stock concentration, we study the subset of investors who remained in the stock market. We find similar results to those of the main text in terms of the signs and significance of all the coefficients. The number of stocks in an investor's portfolio decreases by 0.02, 0.05, and 0.07 in one, three, five years, after having experienced an equity loss that valued 4% their pre-crisis wealth. The stock concentration (*HHI*) of an investor's equity portfolio increases on average by by 1.2%, 4.2%, 5.1% in one, three, five years after the 2008 crisis, for those who experienced a large loss. Furthermore, the magnitude of the coefficients on *Large loss dummy* remains the same across different thresholds of the large loss (compared to the bottom panel of Table 4).

In summary, our findings that investors intend to reduce their risk exposure to equities and increase the concentration of stocks after experiencing a large loss of equity are robust to a more lenient threshold (the median *Total loss%*) for the *Large loss dummy*.

Table A.4: Using a different threshold to define large losses

This table replicates Table 3 and 4 using the median of *Total loss%* as the threshold. *Large loss dummy* equals one if *Total loss%* lies above the median ($Total\,loss\% > 4\%$) and zero otherwise. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation			
Large loss dummy	-2.29***	-0.91***	-1.18***
	(0.05)	(0.29)	(0.34)
Controls	Yes	Yes	Yes
Sample size	344,209	336,623	329,016
Adjusted R^2	0.010	0.007	0.008
Dependent Variable: Stock liquidation N-year after the crisis (Logit)			
Large loss dummy	0.31***	0.27***	0.31***
	(0.04)	(0.04)	(0.03)
Controls	Yes	Yes	Yes
Sample size	344,209	336,623	329,016
Pseudo R^2	0.43	0.22	0.16
Dependent Variable: N-year change in the number of stocks			
Large loss dummy	-0.02***	-0.05***	-0.07***
	(0.001)	(0.003)	(0.003)
Controls	Yes	Yes	Yes
Sample size	292,468	265,278	242,020
Adjusted R^2	0.003	0.006	0.010
Dependent Variable: N-year change in stock concentration			
Large loss dummy	1.16***	4.19***	5.06***
	(0.09)	(0.18)	(0.23)
Controls	Yes	Yes	Yes
Sample size	292,468	265,278	242,020
Adjusted R^2	0.003	0.013	0.016

Appendix E: Formative-year experience

To capture investors' formative-year experience, we calculate the weighted average of past stock market returns when investors were between the ages of 11 and 20, as defined in Equations (5) and (6). We follow Malmendier and Nagel (2011) by introducing a weighting function with parameter λ, which allows the influence of experiences from the distant past to differ from more recent experiences. In the main text, we show results using $\lambda = 1.5$, as Malmendier and Nagel (2011) use $\lambda = 1.5$ for their main analysis. In this section, we conduct a robustness test using $\lambda = 0$, which assigns equal weights to all years, effectively taking the simple moving average of 10-year returns during an investor's formative years.

Figure A.1 illustrates the weighting function (Equation 6), showing the weights $w_{i,t}$ as a function of the time t for two values of $\lambda = 0$ and $\lambda = 1.5$. Figure A.2 depicts the *Formative-year experience (FYE)* variable using these weighting parameters for investors born between 1905 and 1990. We observe a large variation in the formative-year returns. Those whose were born in the 1900s and the late 1950s experienced negative returns, while those born around 1920, 1950, 1960-70, and 1985 experienced largely positive returns. The patterns are consistent regardless of the weighting parameter. Not surprisingly, the formative year returns using $\lambda = 0$ are smoother than those using $\lambda = 1.5$. The average *FYE* is 4.6% and 4.8% for $\lambda = 0$ and $\lambda = 1.5$, respectively.

Table A.5 summarizes the regression results using $\lambda = 0$ to construct *FYE*. The signs and magnitudes of the coefficients for *Large loss dummy* are very similar to those using $\lambda = 1.5$. For example, the coefficients of *Large loss dummy* are all negative and significant in the top panel, where the dependent variables are changes in stock allocation, and in the third panel, where the dependent variables are changes in the number of stocks. They are all positive and significant in the Logit regressions on the likelihood of stock liquidation and in the bottom panel, where the dependent variables are changes in stock concentration. The magnitudes of these coefficients are also comparable to those using $\lambda = 3$. These findings indicate that our previous conclusions that individual investors experiencing a large loss in the financial crisis tend to reduce their stock allocation and increase their stock concentration are robust to controlling for formative-year experience, whether using $\lambda = 0$ or $\lambda = 1.5$ as the weighting parameter.

In addition to personal crisis period, *Formative-year experience* also appears to influence investment decisions. The coefficients for *Formative-year experience* are negative and significant changes in stock allocation and changes in the number of stocks, and are positive and significant for the likelihood of stock liquidation and changes in stock concentration. Although some signs of coefficients for *FYE* differ from those using $\lambda = 1.5$, they do not affect the signs and significance of coefficients for *Large loss dummy*. The interaction term's coefficients are positive and significant for change in stock allocation and stock concentration, indicating that among the investors who experienced large losses

during the financial crisis, those with higher formative-year returns are associated with a larger increase in stock allocation and stock concentration for up to five years after the crisis.

These results support our previous findings that the personal crisis experience and the stock market performance during investors' formative years together have a lasting impact on their equity allocation decision.

Figure A.1: Weighting scheme for FYE

This figure depicts the the weighting scheme for calculating *Formative-year experience*, where the weighting parameter λ takes the value of 0 or 1.5. The weighting function is specified in Equation (6).

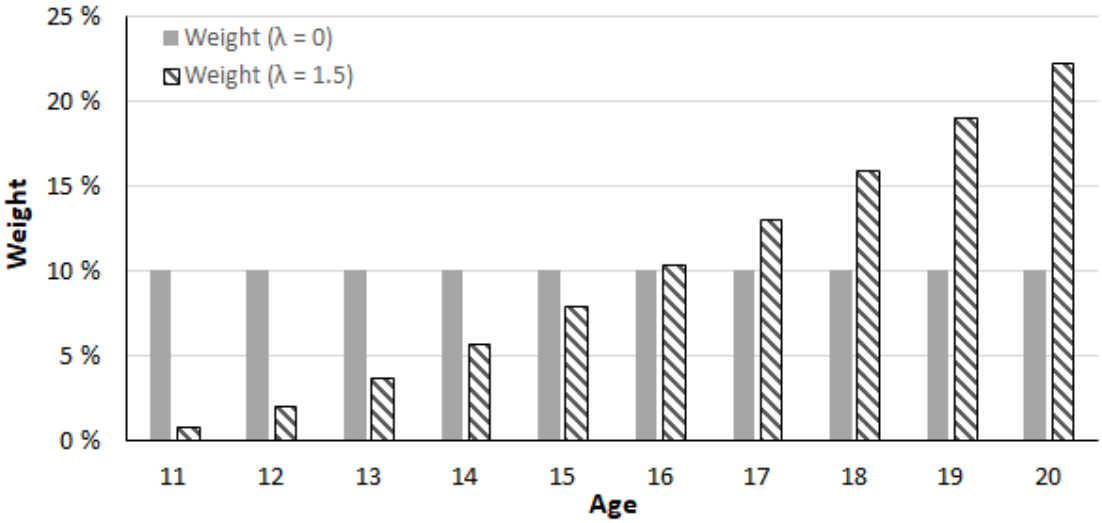


Figure A.2: Formative-year experience (FYE) over time

This figure plots the variable *Formative-year experience* (FYE), defined in Equation (5), where the weighting parameter λ takes the value of 0 or 1.5 (Equation 6).

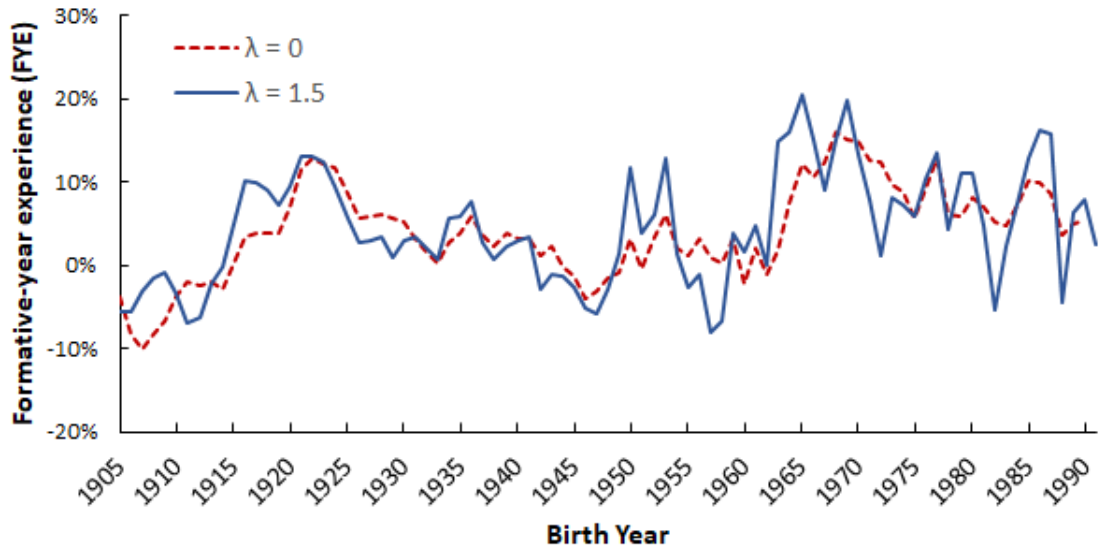


Table A.5: Formative-year experience using $\lambda = 0$

This table reports the results for regressions specified in estimations in equations (12)-(13). Formative-year returns (FYE) is defined in equations (5) and (6) where $\lambda = 0$ is a weighting parameter. The coefficients of control variables are not tabulated, but all the control variables are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Loss Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation						
Large loss dummy	-4.31***	-5.49***	-5.48***	-3.16***	-3.15***	-3.53***
	(0.25)	(0.93)	(0.63)	(0.09)	(0.37)	(0.50)
FYE	-1.67**	7.73***	10.72***	-1.67**	6.33***	8.40***
	(0.71)	(1.84)	(2.43)	(0.71)	(1.28)	(2.07)
Large loss dummy × *FYE*	16.72***	16.00***	19.68**	8.19***	13.38**	18.54***
	(3.73)	(6.49)	(9.21)	(1.93)	(5.93)	(7.25)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Adjusted R^2	0.009	0.011	0.010	0.011	0.010	0.009
Dependent Variable: Stock liquidation N years after the crisis (Logit)						
Large loss dummy	0.59***	0.52***	0.49***	0.46***	0.41***	0.40***
	(0.05)	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)
FYE	0.74***	0.55**	0.36*	0.80***	0.64***	0.42**
	(0.26)	(0.22)	(0.19)	(0.26)	(0.24)	(0.21)
Large loss dummy × *FYE*	0.51	0.74*	0.86**	-0.18	-0.24	-0.04
	(0.53)	(0.40)	(0.39)	(0.40)	(0.36)	(0.32)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Pseudo R^2	0.400	0.195	0.134	0.400	0.195	0.134
Dependent Variable: N-year change in the number of stocks						
Large loss dummy	-0.03***	-0.07***	-0.09***	-0.02***	-0.05***	-0.08***
	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)
FYE	-0.03***	-0.07***	-0.08***	-0.01	-0.04*	-0.04
	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.03)
Large loss dummy × *FYE*	-0.11***	-0.26***	-0.03	-0.11***	-0.20***	-0.24***
	(0.05)	(0.08)	(0.10)	(0.03)	(0.05)	(0.07)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R^2	0.003	0.007	0.010	0.004	0.008	0.013
Dependent Variable: Change in stock concentration index after the crisis experience						
Large loss dummy	1.19***	4.12***	5.12***	0.82***	3.91***	4.66***
	(0.21)	(0.29)	(0.32)	(0.13)	(0.18)	(0.21)
FYE	0.82*	1.97**	0.89	1.11*	3.24***	3.71***
	(0.45)	(0.80)	(0.89)	(0.67)	(1.14)	(1.44)
Large loss dummy × *FYE*	7.69***	11.60***	14.59***	8.31***	12.25***	15.33***
	(2.78)	(3.61)	(5.00)	(1.59)	(2.54)	(3.47)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	292,468	265,278	242,020	292,468	265,278	242,020
Adjusted R^2	0.003	0.008	0.010	0.003	0.012	0.014

Appendix F: Including pre-crisis stock portfolio returns

In addition to the formative-year experience of stock market returns, personal experiences with past stock portfolios can also shape investor's future investment behaviors. For instance, Barber and Odean (2001) find that past trading success leads to overconfidence, particularly among men, which influences their future trading behavior. Malmendier and Nagel (2011) show that individuals who experienced low stock market returns during their formative years tend to be more risk-averse in their investment decisions later in life and that more recent return experiences have stronger effects.

In this section, we construct a variable *Pre-crisis return*, which represents the value-weighted total stock return (both realized and unrealized) for each individual investor based on their tradings and holdings of stock portfolio in 2007. We include *Pre-crisis return* in the baseline regressions (Equations 7-10) and summarize the results in Table A.6, focusing on the coefficients and standard errors of the main variables.

As shown in the table, all coefficients *Large loss dummy* are significant and exhibit the same signs as those based in the main text. The magnitudes of all coefficients are virtually the same after controlling for the pre-crisis stock portfolio returns at the individual investor level. The coefficients for *Pre-crisis return* are largely significant. They are negatively associated with future changes in stock allocation, the likelihood of stock liquidation, and changes in stock concentration, and positively correlated with changes in the number of stocks. These results suggest that investors who experienced high stock portfolio returns before the crisis tend to lower their risk exposure to equities and reduce stock concentration in the years following the crisis.

In summary, these results confirm our previous findings that the personal crisis experience has a lasting impact on investors' equity allocation decision, even after accounting for personal pre-crisis stock portfolio returns.

Table A.6: Including the pre-crisis stock returns

This table replicates Table 3 and 4 by adding *Pre-crisis returns*, which represents the value-weighted total stock return for each individual investor based on their tradings and holdings of stock portfolio in 2007. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Losses Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Dependent Variable: N-year change in stock allocation						
Large loss dummy	-3.56***	-4.39***	-4.55***	-2.87***	-2.08***	-2.29***
	(0.20)	(0.50)	(0.58)	(0.09)	(0.40)	(0.49)
Pre-crisis return	-0.29*	-3.23***	-2.04***	-0.15	-3.27***	-2.06***
	(0.16)	(0.31)	(0.35)	(0.17)	(0.33)	(0.37)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	303,657	296,552	289,378	303,657	296,552	289,378
Adjusted R^2	0.014	0.014	0.014	0.017	0.012	0.016
Dependent Variable: Stock liquidation N-year after the crisis (Logit)						
Large loss dummy	0.76***	0.64***	0.56***	0.55***	0.46***	0.43***
	(0.04)	(0.03)	(0.02)	(0.03)	(0.03)	(0.02)
Pre-crisis return	-0.39***	-0.42***	-0.08*	-0.42	-0.45***	-0.11***
	(0.04)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	303,657	296,552	289,378	303,657	296,552	289,378
Pseudo R^2	0.339	0.183	0.398	0.399	0.183	0.398
Dependent Variable: N-year change in the number of stocks						
Large loss dummy	-0.03***	-0.07***	-0.11***	-0.03***	-0.06***	-0.08***
	(0.002)	(0.004)	(0.005)	(0.002)	(0.003)	(0.004)
Pre-crisis return	0.02***	0.06***	0.08***	-0.02***	0.06***	0.08***
	(0.002)	(0.003)	(0.004)	(0.002)	(0.003)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	303,657	296,552	289,378	303,657	296,552	289,378
Adjusted R^2	0.003	0.007	0.003	0.004	0.008	0.004
Dependent Variable: N-year change in stock concentration						
Large loss dummy	1.16***	4.33***	5.65***	0.93***	4.20***	5.09***
	(0.15)	(0.22)	(0.28)	(0.10)	(0.16)	(0.22)
Pre-crisis return	-0.22	-4.66***	-6.76***	-0.27	-4.93***	-7.06***
	(0.18)	(0.25)	(0.30)	(0.18)	(0.24)	(0.30)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	303,657	296,552	289,378	303,657	296,552	289,378
Adjusted R^2	0.003	0.008	0.002	0.003	0.012	0.003

Appendix G: Full sample analysis for changes in the number of stocks

To capture changes in investors' stock concentration, we analyzed in the main text changes in the number of stocks and stock concentration index for a sub-sample of investors who did not completely liquidated their stock holdings. This focus was necessary because we cannot calculate the stock concentration index for non-stock holders. However, for changes in the number of stocks, which is the difference between the number of unique stocks held at the end of this period and that of the previous period, we do not need to limit our analysis to this sub-sample. For non-stockholders, the number of stocks is simply zero. Thus, in this section, we re-run the baseline regression specified in Equation (9) for the full sample. The results are presented in Table A.7.

The table shows that the coefficients *Large loss dummy* are all negative and statistically significant at 1% level, consistent with the results in Table 4. Regarding the magnitude of the coefficients, they are slightly more negative than those for the sub-sample of investors who remained in the stock market after the crisis. This is not surprising, as those who liquidated all the stocks are now included in the sample, resulting in more drastic drops in the number of stocks from a positive number to zero. In short, our previous finding that personal crisis experience is associated with changes in the number of stocks in investors' portfolio is robust when applied to the full sample.

Table A.7: Changes in the number of stocks for the full sample

This table replicates Table 3 and 4 using the *Total loss%* as an alternative measure for the large losses dummy. This sample includes investors who traded stocks on Oslo Stock Exchange in 2008. The coefficients of control variables are not tabulated, but all the control variables specified in Equation (7) - (10) are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Losses Dummy using Different Thresholds:					
	Above the 90th Percentile			Above the 75th Percentile		
	1-Year	3-Year	5-Year	1-Year	3-Year	5-Year
Large loss dummy	-0.06***	-0.11***	-0.16***	-0.04 ***	-0.09***	-0.13***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	344,209	336,623	329,016	344,209	336,623	329,016
Adjusted R^2	0.003	0.005	0.007	0.004	0.007	0.009

Appendix H: Difference-in-differences estimates

An alternative approach to study the effect of personal crisis experience on stock allocation is to employ difference-in-differences estimations. In this section, we first perform a difference-in-differences analysis using panel data spanning five years before and five years after the crisis. We further conduct a collapsed difference-in-differences analysis, as Bertrand et al. (2004) argue that collapsing the data into pre- and post- periods yields consistent standard errors even when the number of states is small.

To investigate the relationship between personal crisis experience and stock allocation before and after the crisis, we use the panel data for all investors in 2003-2013 (five years before and five years after the crisis). We begin with the following regression:

$$\begin{aligned} StockAllocation_{i,t} = \alpha + \beta_1 \text{Large loss dummy}_i \times Post_{i,t} \\ + \beta_2 \text{Large loss dummy}_i + \beta_3 Post_{i,t} + \gamma X_{i,t-1} + \theta P_{i,t} + \epsilon_i \end{aligned}$$

where i and t denote the individual investor and the year, respectively. The dependent variable $StockAllocation_{i,t}$ is the end-of-year value of stock holdings as a fraction of the end-of-year personal wealth, multiplied by 100. The independent variable *Large loss dummy*, defined in Subsection 2.3.3, is an indicator for personal crisis experience. We consider two settings regarding the inclusion of the crisis year 2008. When including 2008, we define the dummy variable *Post* as equals one if is in 2008 or later and zero otherwise. When 2008 is excluded, *Post* equals one from 2009 onward. The interaction term is thus the parameter of interest. The vector $X_{i,t-1}$ includes the same set of investor wealth-related variables as in the main analysis measured with one- year lag plus the lagged stock allocation to control for the previous level of stock allocation. $P_{i,t}$ is the same set of personal characteristics as in the main analysis, including male, age, married, college degree, and business education.

Unlike the main analysis, where we study different horizons from one to five years after the crisis, in the difference-in-differences analysis, we focus on the averaged yearly change in stock allocation before and after the crisis. For years before 2007, we have limited information on transactions and holdings, thus we can only obtain a coarse measure based on the end-of-year values of stock holdings. Unlike our measure in the main analysis, here we cannot distinguish between active changes in stock allocation and passive changes due to market value fluctuations. We again cluster standard errors along two dimensions, wealth and age cohorts.

Additionally, we perform collapsed difference-in-differences using the following regression:

$$\begin{aligned} \overline{StockAllocation_{i,t}} = \alpha + \beta_1 \text{Large loss dummy}_i \times Post_{i,t} \\ + \beta_2 \text{Large loss dummy}_i + \beta_3 Post_{i,t} + \gamma \overline{X_{i,t-1}} + \theta \overline{P_{i,t}} + \epsilon_i \end{aligned}$$

where i denotes the individual investor; and t consists of two periods, pre- and post- the crisis. The estimation equation is very similar to the previous one except that the dependent variable and controls are collapsed into a "pre" and a "post" period.

Table A.8 presents results for the above two types of difference-in-differences estimations. As shown in the table, all coefficients for the interaction term are negative and significant, indicating that investors tend to allocate less wealth into stocks after having experienced a large equity loss. This negative relation holds for different thresholds of large loss, whether using panel data at yearly frequency or at collapsed at pre- and post-crisis periods, and regardless of whether 2008 is included in the sample. These results are consistent with our previous findings that the personal crisis experience affects investors' equity allocation decision.

Table A.8: Difference-in-differences estimates

This table presents results for difference-in-differences estimations as specified in Appendix H. The coefficients of control variables are not tabulated, but all the control variables are included in the regressions. Standard errors reported in parentheses are clustered at the combination of wealth and age dimensions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Large Loss Dummy using Different Thresholds:			
	Above the 90th Percentile		Above the 75th Percentile	
	Include 2008	Exclude 2008	Include 2008	Exclude 2008
	Difference-in-differences using panel data at yearly frequency			
Large loss dummy × Post	-3.26***	-4.24***	-2.34***	-2.79***
	(0.48)	(0.25)	(0.26)	(0.14)
Large loss dummy	6.46***	6.00***	5.08***	4.73***
	(0.36)	(0.30)	(0.19)	(0.17)
Post	-1.20***	-0.83***	-0.97***	-0.59***
	(0.05)	(0.05)	(0.05)	(0.05)
Controls	Yes	Yes	Yes	Yes
Sample size	3,535,700	3,207,162	3,535,700	3,207,162
Adjusted R^2	0.622	0.643	0.625	0.645
	Collapsed difference-in-differences			
Large loss dummy × Post	-5.27***	-9.26***	-3.96***	-5.81***
	(0.78)	(0.38)	(0.55)	(0.24)
Large loss dummy	11.67***	10.77***	9.35***	8.62***
	(0.93)	(0.70)	(0.51)	(0.37)
Post	-3.14***	-2.06***	-2.62***	-1.50***
	(0.15)	(0.14)	(0.18)	(0.17)
Controls	Yes	Yes	Yes	Yes
Sample size	660,248	657,009	660,248	657,009
Adjusted R^2	0.492	0.548	0.506	0.559